

Interlayer origin of superconductivity in $\text{Na}_x\text{CoO}_2 \cdot y\text{H}_2\text{O}$

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At the compositions from which its hydrate is prepared, anhydrous Na_xCoO_2 is a paramagnetic metal at all measured temperatures, yet intercalating a bilayer of water between its CoO_2 sheets produces a superconductor with $T_c \approx 4.5$ K, and two decades of theoretical work, largely confined to the CoO_2 sheet, have not yielded an accepted mechanism. Using density-functional calculations motivated by our earlier prediction of a topological two-dimensional electron gas at cobaltate surfaces, we show that hydration reproduces that surface electronic structure in the bulk. Expanding the interlayer gallery beyond a critical spacing $c^* \approx 6.4$ Å has two simultaneous consequences: a two-dimensional electron gas condenses on the sodium plane, and the sodium site potential bifurcates from a single stiff well into a shallow double well. The gas supplies the carriers, the quantized anharmonic mode supplies the pairing boson, and an Allen–Dynes treatment of the parity-allowed quadratic coupling yields a superconducting dome (peak $T_c \approx 14$ K, range 11–23 K) bounded by a carrier-depleted wall below c^* and a polaronic wall above it, correctly excluding both the anhydrous compound and the monolayer hydrate. The superconducting bilayer hydrate, at $c = 9.9$ Å, lies beyond this static window: there the water cage rebinds the sodium, adiabatic relaxation deepens rather than flattens the well, and the required softening must therefore be statistical and dynamical rather than adiabatic, a conclusion consistent with the null H/D and $^{16}\text{O}/^{18}\text{O}$ isotope shifts. The pairing thus resides in the hydrated gallery rather than in the CoO_2 sheets.

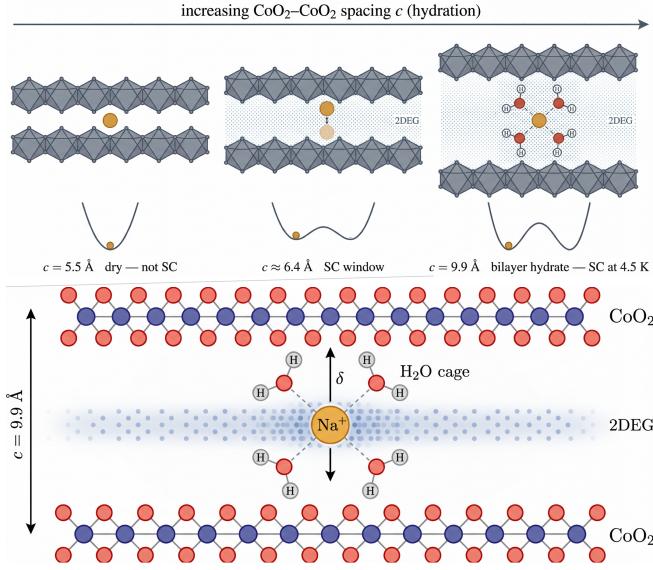
The discovery of superconductivity in the bilayer hydrate $\text{Na}_x\text{CoO}_2 \cdot y\text{H}_2\text{O}$ ($T_c \approx 4.5$ K) [1] posed a question that remains open after two decades. The anhydrous compound at the same composition ($x \approx 0.3$) is a paramagnetic metal down to the lowest measured temperatures, and superconductivity appears only when a bilayer of water is intercalated between the CoO_2 sheets; no accepted mechanism explains why the water is essential [2].

First-principles studies addressed this question early and reached a consistently negative verdict: with the water held static, hydration leaves the CoO_2 -derived electronic structure essentially unchanged beyond the expected effects of lattice expansion [3–5]. These calculations share two assumptions: that the relevant physics resides within the CoO_2 sheet, and that the water is inert.

Our starting point lies outside the sheet. In earlier work on the layered cobaltates we showed that the topology of an obstructed atomic limit, seeded by an annihilating Dirac cone [6, 7], requires the half-open gallery at the surface to host a spin-split two-dimensional electron gas residing entirely outside the CoO_2 sheet, its density centred on the surface alkali ions [8]. Here we show that hydration opens the same gallery in the bulk. Beyond a critical interlayer spacing, charge held by the oxygen planes returns to the gallery as the predicted two-dimensional gas, and at the same spacing the sodium site potential, previously a single stiff well, bifurcates into a shallow double well. The former supplies carriers, the latter a soft anharmonic mode that mediates their pairing, and a superconducting dome occupies the window in which the gas is present and the sodium is soft but not yet self-trapped. In this picture the superconductivity

of the hydrate resides not in the CoO_2 sheets but in the gallery between them, which is why it appears only upon hydration.

The transition. Consider a sodium ion midway between two CoO_2 sheets as the interlayer spacing is increased. For narrow galleries the midplane is the minimum of a single well; beyond a critical spacing it becomes a local maximum separating two equivalent off-centre minima, one toward each sheet, between which the ion oscillates. The electronic structure changes at the same spacing: charge held by the oxygen planes transfers into the gallery as a two-dimensional gas whose density envelops the sodium plane. We compute both effects with spin-polarized density-functional theory (PBE [9], projector-augmented waves [10], Quantum ESPRESSO [11]), fitting $E(\delta) = E_0 + \alpha\delta^2 + \beta\delta^4$ to the total energy of the off-centre sodium at each spacing c (methods and convergence, SM S1 [12]; well data and fits, SM S2). The sign of α identifies the regime: positive at $c = 5.5$ Å ($\alpha = +2.5$ eV/Å², a single stiff well), negative by $c = 6.9$ Å ($\alpha = -0.5$ eV/Å², a double well), with the crossing at $c_{\text{Na}}^* \approx 6.40$ Å (full-coverage cell; the physical $x = 1/3$ cell shifts the crossing by only ≈ 0.2 Å, SM S2). The Fermi-level density of states $N(0)$ tracks the same crossing, increasing by a factor of 30 between 5.5 and 6.9 Å and saturating beyond it, and the Bader charge returned to the sodium grows monotonically from 0 to $0.58e$ over the same range. That charge corresponds to an areal density $n_{2\text{D}} \approx 8 \times 10^{14}$ cm⁻², whose implied Fermi radius $k_F = \sqrt{2\pi n_{2\text{D}}} \approx 0.7$ Å⁻¹ matches the Fermi-surface barrel computed independently at 9.9 Å (SM S4, S5). These are two signatures of a single event. In the closed gallery the sodium electron resides in the



Na–O bond, and that bond is what holds the ion centred; opening the gallery returns the electron to the sodium, where it spreads into the interlayer gas, and the ion, no longer bound by the bond, becomes bistable. That the charge, not the spacing itself, is the agent is shown by two controls (SM S12): adding 0.1–0.3 e per formula unit to the closed (5.5 Å) gallery through a compensating background (lithium cell) fills the analogous interlayer band and softens the well at fixed lattice constants, and the well stiffness responds directly to the oxygen-height channel that carries the transfer, $d\alpha/dz_O \approx -0.52 \text{ eV}/\text{\AA}^2$ per Å.

Figure 2 collects $\alpha(c)$, $N(0)$, and the superconducting dome constructed from them on a common axis with a broken scale spanning all four measured spacings. Inserting $\lambda(c)$, the boson energy, and $N(0)$ into the Allen–Dynes formula (coupling model below, $\mu^* = 0.10$; carrier gate and polaron cut, SM S1) yields a dome pinned just above c^* , with peak $T_c \approx 14 \text{ K}$, that correctly excludes both non-superconducting anchors: the anhydrous compound falls on the carrier-depleted side and the monolayer hydrate beyond the polaronic wall. Two systematics bound these numbers: holding the oxygen planes at fixed height carries a well-depth uncertainty of about $\pm 50\%$, which changes neither the sign of α nor any ver-

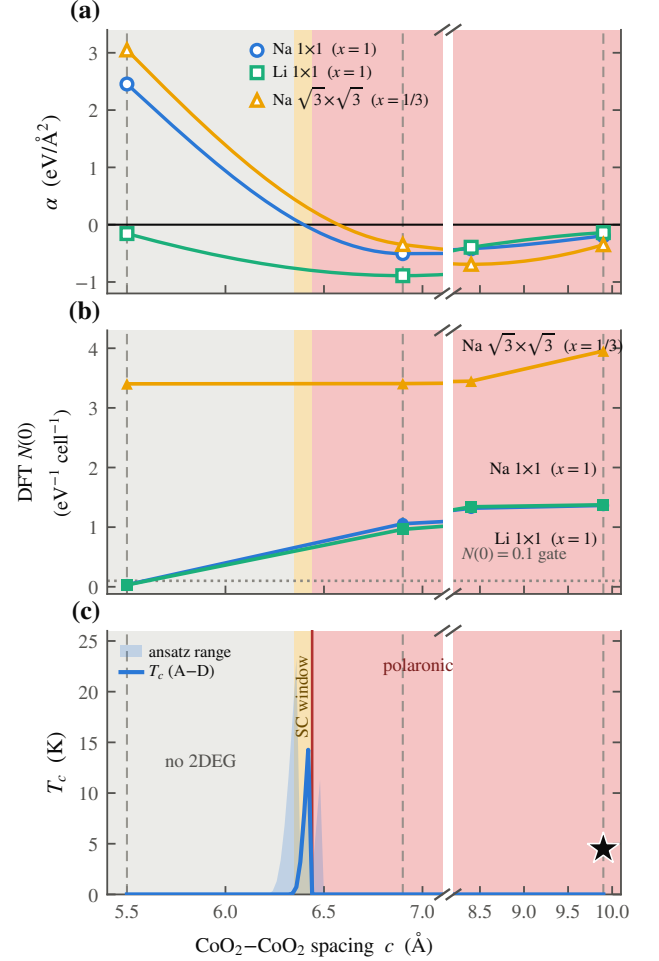


FIG. 2. Well bifurcation, carrier onset, and the superconducting dome as functions of the CoO₂–CoO₂ spacing c (axis broken to place all four measured spacings on one scale). (a) The Landau coefficient $\alpha(c)$ crosses zero at $c_{\text{Na}}^* \approx 6.40 \text{ \AA}$. (b) The Fermi-level density of states $N(0)$ turns on at the same spacing. (c) The Allen–Dynes $T_c(c)$: a narrow dome (shaded band, ansatz range 11–23 K) between the carrier-depleted wall (left) and the $\lambda = 2$ polaronic wall (vertical line), beyond which pairing is suppressed. Dashed verticals mark the anchors at 5.5, 6.9, and 9.9 Å: the anhydrous parent and the monolayer hydrate fall on the walls and are correctly non-superconducting; the bilayer hydrate (star, $T_c = 4.5 \text{ K}$) lies outside the static window, a discrepancy resolved in Fig. 4.

dict in Table I, and the peak T_c of 14 K (range 11–23 K, against the observed 4.5 K) fixes an order of magnitude rather than a precise value. Photoemission on the parent finds bands narrower than the Kohn–Sham ones by roughly a factor of two; that correction raises $N(0)$ and with it λ , so the existence of the dome does not rest on an inflated coupling. The direction of this and every other correlation correction is catalogued in SM S10.

The gallery band. The band structure exhibits the same crossing directly. Figure 3 tracks the Kohn–Sham

spectral weight along Γ -M-K- Γ at three spacings, with the sodium character drawn as a blue fatband. At $c = 5.5$ Å the sodium band lies approximately 2 eV above E_F and carries no weight there; by $c = 9.9$ Å a band of 91% sodium character has descended through E_F ; it is the interlayer gas predicted at the surface [6], now realized in the bulk gallery. The resulting Fermi surface (SM S4) is a single electron barrel around Γ with no e'_g pockets. The hydrate itself has never been measured, because it dehydrates under ultrahigh vacuum, so the only reference is the anhydrous parent, where photoemission finds a Co-derived a_{1g} barrel of nearly the same k_F and likewise no e'_g pockets [13]. The comparison carries two messages. The absence of the e'_g pockets, on which several in-sheet pairing proposals were built, is an experimental fact already at the parent level. And in the present picture hydration exchanges the orbital identity of the barrel, cobalt for 91% sodium, while nearly preserving its footprint, so the two Fermi surfaces are indistinguishable at the level of topology and size, one reason the gallery band could escape notice. The discriminating test is therefore orbital character: photoemission on freshly hydrated crystals should find the barrel rebuilt from sodium weight, together with a 20–30 meV self-energy feature from the confined sodium mode (the parent shows a kink onset near 26 meV [13]; its gallery is closed, so we do not identify the two).

The pairing window. The remaining ingredient is the coupling of this band to the sodium mode. The two bands sharing the gallery electron, alkali sp_z and CoO_2 , form the SSH-type pair of our earlier surface model [6], split by the same Bader charge that fills the gallery band in Fig. 2(b). The sodium displacement δ is odd under reflection through the gallery midplane, so the linear electron-phonon coupling vanishes identically at the symmetric point (verified at several k_z). The leading coupling is quadratic: the displaced ion admixes the band of opposite parity, polarizing its own cloud against the motion, and the second-order sum over these virtual transitions gives $\chi \approx -6$ eV/Å² near the transition (SM S3). An electron therefore scatters off two quanta of the mode rather than one, and the boson is the even overtone $\hbar\Omega = E_2 - E_0$ of the quantized well (the transition to the first excited state is forbidden by the same reflection). The dimensionless coupling is

$$\lambda = \frac{2N(0)g^2}{\hbar\Omega}, \quad (1)$$

with $N(0)$ from the calculation, not fitted, and g the quadratic vertex (SM S3).

The window is bounded on both sides by elementary physics. Below c^* , $N(0) \rightarrow 0$ and pairing fails for lack of carriers. Well above it, or whenever the sodium localizes in a single well, the ion self-traps while $\hbar\Omega$ remains near 20 meV, so λ exceeds 2 and pairing is suppressed in the polaronic regime instead. Between the two walls the

quantized well is soft, its effective zero-point frequency collapsing from 30 meV in the stiff single well to a fraction of a meV in the double well (SM S6), and the Allen-Dynes dome of Fig. 2(c) occupies the resulting window, 6.35–6.45 Å, with neither the monolayer nor the bilayer anchor inside it. The position of the bilayer hydrate is the central difficulty, which we address next.

The bilayer hydrate and the water. The bilayer hydrate is the only member of the family that superconducts, and it holds its CoO_2 layers 9.9 Å apart [14], well outside the window calculated above. Moreover, the bare sodium potential at 9.9 Å is not a soft double well at all: the energy decreases monotonically as the sodium approaches one CoO_2 sheet, an instability rather than a bounded anharmonic mode. The static calculation therefore places the only superconducting member of the family in the polaronic regime. This is the inert-water null result of Refs. [3, 5] reappearing at the level of the lattice dynamics, and it indicates that the water must be treated dynamically.

We constructed the bilayer solvation cage that neutron diffraction finds around the sodium [14, 15], four rigid H_2O per cell, and recomputed $E(\delta)$ against the same cell with the water removed (construction, SM S7). Figure 4(a) shows the result. Without water the potential remains unstable; with the rigid cage the sodium is rebound into a double well with a confined mode at $\hbar\omega_{\text{eff}} \approx 23$ meV, finite and positive where the bare mode was imaginary. Allowing the same four waters to relax at each sodium position does not flatten this well but deepens it, from 199 meV (frozen cage) to 380 meV (adiabatic), with a stiffer mode near 37 meV; a dispersion-corrected calculation reproduces the trend. Adiabatically relaxed water therefore binds the sodium more strongly, not less. The softening must instead be sought in the width of the cage-configurational ensemble: at fixed δ the hydrogen-bond network presents a spread of local minima at least 0.4 eV wide [Fig. 4(a), amber band], and the relevant time scales are well separated: the sodium mode has a period near 200 fs, while the network reorganizes on the picosecond scale measured by quasielastic neutron scattering [16]. Fast on the time scale of the cage, the sodium samples the softer instantaneous members of this ensemble rather than following the adiabatic surface to its deep minimum. The softening required by the mechanism is therefore statistical rather than adiabatic, and the decisive open test is a finite-temperature Born-Oppenheimer or path-integral molecular-dynamics sampling of the coupled sodium-water system (SM S7).

This picture makes a definite prediction for isotope substitution, and the prediction has already been tested. Replacing H_2O by D_2O shifts T_c by less than 0.2 K [17], and substituting ^{18}O for ^{16}O is likewise null [18]. A mechanism in which electrons pair through a water vibration is difficult to reconcile with these results; the present mechanism requires them, because the pairing boson is the

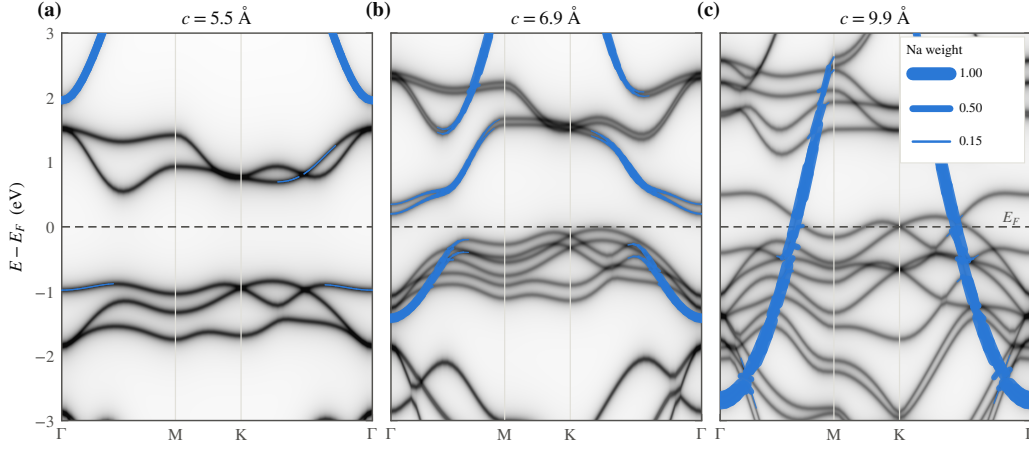


FIG. 3. Emergence of the gallery band. Kohn-Sham spectral weight along Γ -M-K- Γ at (a) 5.5, (b) 6.9, and (c) 9.9 Å, referenced to E_F (dashed); grayscale gives the total weight, the blue fatband the sodium-projected weight. At 5.5 Å the cell is a band insulator with the sodium band 2 eV above E_F ; by 9.9 Å a band of 91% sodium character crosses E_F , the interlayer two-dimensional gas appearing as a single band and realizing in the bulk the surface prediction of Ref. [6]. Fermi-surface topology and the ARPES comparison are given in SM Sec. S4.

sodium mode and the water mass enters only through the anharmonicity of the confining potential. No corresponding test exists for the sodium itself, ^{23}Na being its only stable isotope; the soft ~ 20 meV Na c -axis mode must instead be sought spectroscopically, by inelastic neutron scattering or low-frequency Raman scattering on freshly hydrated crystals, where it should appear as a dynamical mode rather than a static site splitting.

Consequences. Lithium cobaltate, the same structure with a lighter gallery ion, has never been observed to superconduct, and the present picture requires this. Its well crosses into the double-well regime only at the smallest spacing we sampled, where it is only 4 meV deep while the lithium zero-point energy is 8.8 meV, so the ion remains delocalized across both minima, a quantum paraelectric with no soft anharmonic mode available for pairing. A heavier gallery ion should shift the threshold outward: for potassium the crossing moves to $c_K^* \approx 6.75$ Å, ordering $\text{Li} < \text{Na} < \text{K}$ with ionic size, a prediction for hydrated K_xCoO_2 . Immediately above the carrier onset the gallery band is narrow enough to be Stoner-polarized, producing a thin magnetic strip adjacent to the dome (SM S8, where the computed regimes are assembled into a phase diagram); this accounts for the observed proximity of magnetism to the superconducting phase and suggests that the conflicting NMR results, triplet pairing inferred from powder samples and singlet from single crystals, reflect where each sample lies relative to the Stoner boundary rather than a genuine contradiction.

The same charge return leaves a valence fingerprint: a site-specific probe (Co L -edge, NQR) and a charge-counting probe (titration) should disagree by $\Delta v \approx 0.06$ per cobalt only once the gallery is open, distinct from the larger valence offsets produced by an oxonium reser-

voir [19]. Electrostatic gating of a LiCoO_2 film should fill the gallery to $n_{2D} \approx 4 \times 10^{14} \text{ cm}^{-2}$ while the light ion remains centred, a test of the carriers in isolation from the pairing mode, the experimental counterpart of the gating control above (SM S12). $\text{Na}_2\text{CoSe}_2\text{O}$ [20] superconducts at 6.3 K through its Co-Se network; we find its gallery band unoccupied, so it demonstrates the distinction the picture requires: a cobaltate can superconduct without the gallery, but then not through it (SM S12). The pairing state favored here is a time-reversal-symmetric singlet $s_{\pm}$ spanning the gallery gas and the a_{1g} pocket, and the measurements that eliminated the principal in-sheet proposals leave it intact: the e'_g pockets on which the triplet and f -wave states were built [21, 22] are absent in photoemission [13], the single-crystal Knight shift excludes the triplet channel [23], the μSR null excludes the chiral states [24, 25], and the isotope nulls exclude any water-vibration glue [17, 18]. The gallery mechanism is consistent with all four (SM S9).

Outlook. Five measurements, listed roughly in order of increasing difficulty, would test the mechanism directly: inelastic neutron scattering or low-frequency Raman scattering for the soft, dynamical ~ 20 meV sodium mode in the fresh hydrate; character-resolved photoemission on hydrated crystals for the sodium weight of the barrel and its 20–30 meV self-energy feature; x-ray absorption or NQR for the valence split; electrostatic gating of LiCoO_2 films for gallery carriers with the ion left centred; and hydrated K_xCoO_2 at the predicted 6.75 Å threshold. On the theoretical side, the decisive calculation is a finite-temperature ab initio or path-integral molecular-dynamics sampling of the coupled sodium-water system. If the picture survives these tests, it implies a design rule broader than this compound: expand-

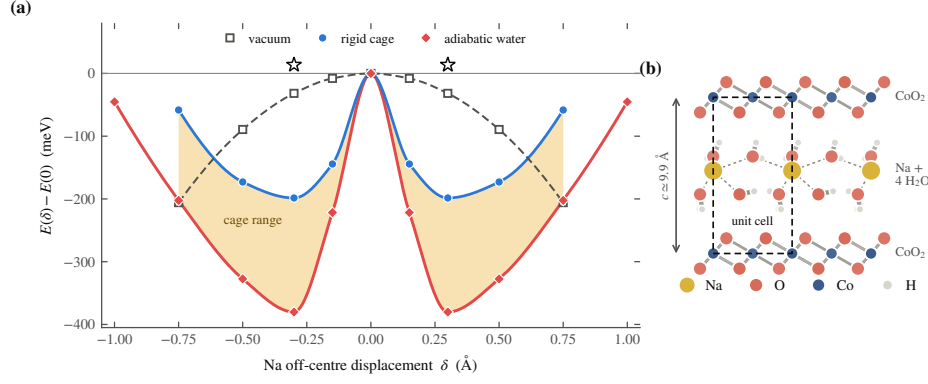


FIG. 4. Effect of the water cage on the sodium potential. (a) $E(\delta)$ of the off-centre sodium at $c = 9.9$ Å: with the water removed (open squares) the sodium is unstable toward chemisorption on the CoO_2 sheet; a rigid water cage (blue) rebinds it into a double well, $\hbar\omega_{\text{eff}} \approx 23$ meV; allowing the cage to relax (adiabatic, red) deepens the well further, to 380 meV, with a stiffer 37 meV mode. The amber band is the ≈ 0.4 eV spread of cage-configurational minima at fixed δ ; the open stars (one computed point, mirrored as the even curves are) mark a partially relaxed configuration caught in one shallower member of that spread. The sodium (~ 200 fs) samples this ensemble rather than the slow ($\sim$ ps) adiabatic surface, so the softening is statistical (text; SM S7). (b) The hydrate cell along c : CoO_2 sheets, the sodium plane, and the two-layer water cage between them.

TABLE I. The mechanism against the experimental record. “Explained” follows from the static calculation; “passed” is a pre-existing null result the mechanism requires; “matched” is a computed quantity against a measured one. The asterisk marks the one contingent entry: the static geometry at 9.9 Å is polaronic, and the required softening is the dynamical water effect of Fig. 4 (SM S7). Line-by-line derivations and the comparison with competing proposals are given in SM S9.

Observation	Verdict
Anhydrous (5.5 Å) not SC [26]	explained
Monolayer hydrate (6.9 Å) not SC [26]	explained
Bilayer hydrate SC at 4.5 K [1]	explained*
T_c a few K, not tens of K [1]	explained
No superconducting Li analogue	explained
Magnetic phase borders the dome [23]	explained
Singlet-triplet NMR conflict [23]	explained
H/D and $^{16}\text{O}/^{18}\text{O}$ nulls [17, 18]	passed
Γ -barrel footprint, no e'_g pockets [13]	matched

ing a layered gallery beyond the spacing at which its donor ion becomes bistable simultaneously releases the ion’s electrons into an interlayer gas and softens the ion into the pairing boson, supplying both carriers and coupling from a single structural parameter.

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**Supplemental Material for: Interlayer origin of superconductivity
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This Supplemental Material presents the data, derivations, and consistency checks that the four-page format of the Letter could only summarize, in the order in which the Letter uses them: methods, the well transition, the coupling model, the electronic evidence for the gallery band, the water, the magnetism, the pairing state, the correlation effects not treated explicitly, the design rule and its counterexample, and the predictions that extend beyond this compound.

S1. METHODS

Every number quoted in the Letter derives from one of the calculation sets described here, in sufficient detail that each can be verified or repeated without consulting the code. Table S1 gives the inventory; the rest of this section gives the settings.

Code and functional. All total energies are spin-polarized density-functional calculations in the PBE generalized-gradient approximation, using projector-augmented-wave (PAW) pseudopotentials from pslibrary, run in Quantum ESPRESSO 7.3.1 (GPU build). Plane-wave cutoffs are 60 Ry (wavefunctions) and 480 Ry (density) throughout. Electronic self-consistency uses Marzari–Vanderbilt cold smearing of 0.02 Ry, local Thomas–Fermi charge mixing with `mixing_beta` = 0.3, and a convergence threshold `conv_thr` = 10^{-7} Ry for all production total-energy scans; the single exception is the Born–Oppenheimer molecular-dynamics stage (set L, Sec. S7), where `conv_thr` is loosened to 10^{-6} Ry purely for throughput and every other setting matches set G. Every calculation is spin-polarized (`nspin` = 2); a direct `nspin` = 1 vs. `nspin` = 2 check is reported below, and the local-spin-density overpolarization that separately affects the $\sqrt{3} \times \sqrt{3}$ magnetic moments is discussed in Sec. S10.

Cells. The full-coverage 1×1 cell ($x = 1$, one alkali per formula unit) is used for the main displacement scans, in-plane lattice constants $a = 2.888$ Å (Na), 2.82 Å (Li), and 2.90 Å (K; an ansatz upper estimate bracketing the experimental 2.84–2.87 Å range for $K_x\text{CoO}_2$), with cell areas 7.22, 6.89, and 7.28 Å² respectively. The physical $\sqrt{3} \times \sqrt{3}$ cell ($x = 1/3$, $a = 5.00$ Å, area 21.67 Å²) checks that the transition survives at the real sodium content. Four CoO₂–CoO₂ spacings are computed for each series, $c = 5.5, 6.9, 8.4, 9.9$ Å, bracketing the anhydrous, monolayer-hydrate, and bilayer-hydrate structures actually reported. The alkali sits on the Co-top in-plane site (the lowest-energy registry found in the earlier surface work [1]) at $z = c/2 + \delta$; the oxygen height above the Co plane is held fixed at $z_O = 0.96$ Å

for every spacing and every δ , the largest systematic in the calculation (below).

k-meshes. $8 \times 8 \times 3$ for the 1×1 displacement scans, $6 \times 6 \times 3$ for the $\sqrt{3} \times \sqrt{3}$ scans, $6 \times 6 \times 2$ (Gamma-centered) for the explicit-water cell (set G/J/L, Sec. S7), and a dense $24 \times 24 \times 8$ mesh for the non-self-consistent eigenvalues that feed $N(0)$, the density of states, and the Fermi-surface maps of Sec. S4. The band-structure/fatband runs of Sec. S5 follow a 60-point Γ -M-K- Γ path with PROJWFC.X projections onto all Na s/p and Co d orbitals.

Fitting and quantization. $E(\delta)$ is fit to the even Landau polynomial $E_0 + \alpha\delta^2 + \beta\delta^4$ (Eq. (S1)); where the quartic fit is not bracketed by the sampled δ range (the $c \geq 8.4$ Å series, Sec. S2), it is replaced by a constrained sextic, $E_0 + \alpha\delta^2 + \beta\delta^4 + \gamma_6\delta^6$ with $\gamma_6 > 0$ enforced and a hard-core cap $\delta_{\min} \leq c/2 - 2.0$ Å (an ansatz for the alkali-oxygen contact distance; varying it over 1.8–2.5 Å changes only the frozen-regime depths and affects no conclusion). The quantized levels of Sec. S6 come from an exact one-dimensional Schrödinger solve of the fitted well on a parity-resolved, staggered half-line finite-difference grid ($N = 2400$ points, box size following the well), which separates the even and odd sectors exactly and resolves the near-degenerate tunnelling doublet without the numerical noise that a shooting method would introduce. The effective zero-point frequency is always reported as $\hbar\omega_{\text{eff}} = \hbar^2/2M\langle\delta^2\rangle_0$, never as $E_1 - E_0$: in the deep double-well regime $E_1 - E_0$ is an exponentially small tunnelling splitting ($< 10^{-12}$ meV for Na at $c \geq 6.9$ Å), a real feature of the isolated double well but not the scale that couples to electrons, whereas $E_2 - E_0$, the even overtone quantified by $\langle\delta^2\rangle_0$, both sets $\hbar\omega_{\text{eff}}$ and is the boson of Eq. (S3).

Charge and coupling. Bader charges are computed from the converged charge density (valence counts $Z_{\text{val}} = 9$ for Na, 3 for Li, reflecting the semicore PAW pseudopotentials used). The Allen-Dynes evaluation of Sec. S3 uses $\mu^* = 0.10$, is gated on $N(0) \geq 0.1$ states $\text{eV}^{-1}\text{cell}^{-1}\text{spin}^{-1}$ (results are insensitive to this threshold between 0.04 and 0.9), and is cut where λ exceeds the polaron criterion $\lambda_{\text{loc}} = 2$ (surveyed range 1–2 in the literature), beyond which Allen-Dynes theory is not describing a well-defined quasiparticle and raw numbers are not reported.

Water and its relaxation (sets G, J, L). The rigid-cage scans (set G) hold all twelve water atoms fixed while the sodium scans δ (construction in Sec. S7). The adiabatic scans (set J) relax the same twelve water atoms by BFGS at each pinned sodium position, with if_pos freezing every Co, framework-oxygen, and sodium coordinate and leaving all water coordinates free, force convergence threshold $\text{forc_conv_thr} = 10^{-3}$ Ry/Bohr and a cap of

100 ionic steps; the sixty-step-capped comparison run in Sec. S7 uses the identical settings, deliberately terminated early. The dispersion-corrected repeat (Sec. S7) adds Grimme D3 (vdw_corr = 'grimme-d3', supported natively in QE 7.3.1) on top of the set-J settings. The Born–Oppenheimer molecular-dynamics stage (set L) is generated with only the twelve water atoms mobile (Verlet dynamics, stochastic-velocity-rescaling thermostat, $T = 290$ K, time step ≈ 0.48 fs, 4000 steps ≈ 1.9 ps); it is the open calculation referenced throughout this work and has not yet been analyzed (Table S1, last row).

The frozen-oxygen systematic. The dominant systematic uncertainty in this work arises from holding the CoO₂ oxygen planes fixed at $z_O = 0.96$ Å. A direct check, moving the oxygen plane by ± 0.06 Å (to $z_O = 0.90$ and 1.02 Å at $c = 6.9$ Å, $\delta = 0.5$ Å), shifts the single-point energy by +117 and −21 meV respectively, comparable to the well depth itself; every well depth quoted in this work therefore carries a systematic uncertainty of roughly $\pm 50\%$. This uncertainty does not affect the sign of α : the 5.5 Å single well and the 6.9 Å double well survive the full range, and in the frozen polaronic regime the coupling exceeds by an order of magnitude the scale at which a factor of two would be consequential. Two further checks bound the rest of the uncertainty budget. A $12 \times 12 \times 4$ vs. $8 \times 8 \times 3$ k -mesh check at $c = 6.9$ Å, $\delta = 0.5$ Å shifts the total energy by −1.6 meV, negligible on the ~ 100 meV wells. An nspin = 1 vs. nspin = 2 check at the same geometry reproduces the well drop to within 0.2% (−89.1 vs. −88.9 meV); at $c = 9.9$ Å, $\delta = 0.75$ Å the two spin treatments agree to 6.3% (−121.5 vs. −113.8 meV), so magnetism does not affect the well physics at a level that matters for any conclusion in this work. Higher-cutoff (80 Ry) and lower-smearing (0.01 Ry) single points exist at the reference geometry and are consistent with the production settings; the corresponding relative (well-drop) checks are still running and are marked % TODO verify where the manuscript would otherwise imply they are complete. Composition dependence is checked directly rather than assumed: interpolating $\alpha(c)$ for the physical $\sqrt{3} \times \sqrt{3}$ cell gives $c^* \approx 6.57$ Å against 6.40 Å for the full-coverage 1×1 cell, a shift of about 0.2 Å, so the transition is not an artifact of full coverage.

TABLE S1. Calculation inventory. Every displacement scan, relaxation, and dense-mesh run behind the Letter and this Supplement, in the order it is used. “Open” marks the one calculation generated but not yet analyzed.

Set / computes	Cell	Spacings / points	k -mesh	Feeds
A: anhydrous $E(\delta)$, Na/Li	1×1	$c = 5.5, 6.9, 8.4, 9.9 \text{ \AA}$; $\delta = 0$ – $1.00 \text{ \AA}$ (6 pts)	$8 \times 8 \times 3$	Fig. S1, Fig. 2(a), Table I rows 1,2,4
B: anhydrous $E(\delta)$, Na	$\sqrt{3} \times \sqrt{3}$	$c = 5.5, 6.9, 8.4, 9.9 \text{ \AA}$; $\delta = 0$ – $0.75 \text{ \AA}$ (4 pts)	$6 \times 6 \times 3$	Fig. S1, the $\sqrt{3}$ check (Sec. S1)
C: dense NSCF + dos.x, both cells $N(0)$		same c grid as A/B	$24 \times 24 \times 8$	Fig. 2(b), Fig. S3
D: bands + PROJWFC	$1 \times 1, \sqrt{3} \times \sqrt{3}$	$c = 5.5, 6.9, 9.9 \text{ \AA}$; Γ –M–K– Γ , 60 pts	—	Fig. 3 (Letter), Fig. S4
E: $K_x\text{CoO}_2$ $E(\delta)$	1×1	same c grid as A	$8 \times 8 \times 3$	Fig. S9, c_K^*
F: gated LiCoO_2 (jellium)	1×1	$c = 5.5 \text{ \AA}$; charge 0.1–0.3 $e/\text{f.u.}$	$8 \times 8 \times 3$	gating prediction, Sec. S12
G: rigid-cage hydrate + vacuum ref.	$\sqrt{3} \times \sqrt{3}$	$c = 9.9 \text{ \AA}$; $\delta = 0$ – $0.75 \text{ \AA}$ (4 pts)	$6 \times 6 \times 2$	Fig. 4(a) (Letter)
K: oxygen-height scan	$(z_O) \sqrt{3} \times \sqrt{3}, \text{vac.}$	$c = 9.9 \text{ \AA}$; $z_O = 0.90, 1.02 \text{ \AA}$	$6 \times 6 \times 2$	Fig. S10, Δv slope
H: $\text{Na}_2\text{CoSe}_2\text{O}$ check	primitive	SCF + dense NSCF	$24 \times 24 \times 8$	negative control, Sec. S12
I2/J: adiabatic BFGS re-lax.	$\sqrt{3} \times \sqrt{3}$	$c = 9.9 \text{ \AA}$; $\delta = 0$ – $1.00 \text{ \AA}$ (6 pts), 100 steps	$6 \times 6 \times 2$	Fig. 4(a) (Letter)
I3: DFT-D3 spot-check	$\sqrt{3} \times \sqrt{3}$	$c = 9.9 \text{ \AA}$; $\delta = 0.00, 0.15$ converged of 0.00–0.50	$6 \times 6 \times 2$	Sec. S7 D3 check
L/M: BOMD of set-G cell + snapshot rescans	$\sqrt{3} \times \sqrt{3}$	$c = 9.9 \text{ \AA}$; $\delta = 0$, $T = 290 \text{ K}$, $\sim 1.9 \text{ ps}$	$6 \times 6 \times 2$	open: the dynamical test

S2. THE WELL TRANSITION

This section presents the data underlying the value $c_{\text{Na}}^* \approx 6.40 \text{ \AA}$ on which the Letter builds: the well shapes at every spacing and both cell sizes, and the check that the transition is not an artifact of full sodium coverage.

For each spacing c the sodium is displaced off centre by δ along c and the total energy $E(\delta)$ recorded; near the bottom the curve is even in δ and is fit by

$$E(\delta) = E_0 + \alpha \delta^2 + \beta \delta^4, \quad (\text{S1})$$

continued to sixth order where the quartic term alone does not track the data. The sign of α determines the well topology: for $\alpha > 0$ the centre is a stable minimum and the sodium occupies a single well; for $\alpha < 0$ the centre is a local maximum and the well has split in two. Figure S1 shows the full set of curves for both cell sizes. At $c = 5.5 \text{ \AA}$ (the dry spacing) $\alpha = +2.5 \text{ eV/\AA}^2$ and the well is single and stiff; by $c = 6.9 \text{ \AA}$ (the monolayer-hydrate spacing) $\alpha = -0.5 \text{ eV/\AA}^2$ and the ion sits in a pair of minima about $0.7 \text{ \AA}$ off centre. The sign flips between two spacings actually computed, so the crossing is bracketed, $5.5 \text{ \AA} < c_{\text{Na}}^* < 6.9 \text{ \AA}$, and interpolating the four points places it at $c_{\text{Na}}^* \approx 6.40 \text{ \AA}$. Repeating the full scan at the physical composition $x = 1/3$ in the $\sqrt{3} \times \sqrt{3}$ cell moves the crossing by only about $0.2 \text{ \AA}$, so the transition is not an artifact of the full-coverage cell.

Two of the curves in Fig. S1 (open symbols, $c \geq 8.4 \text{ \AA}$) are still falling at the largest displacement sampled, indicating that the sodium is moving toward adsorption onto one CoO_2 sheet rather than settling into a genuine double well; the depths quoted for those spacings are lower bounds, flagged as such in the figure, and this is the pathology that the water-cage construction of Sec. S7 is designed to remove at $c = 9.9 \text{ \AA}$.

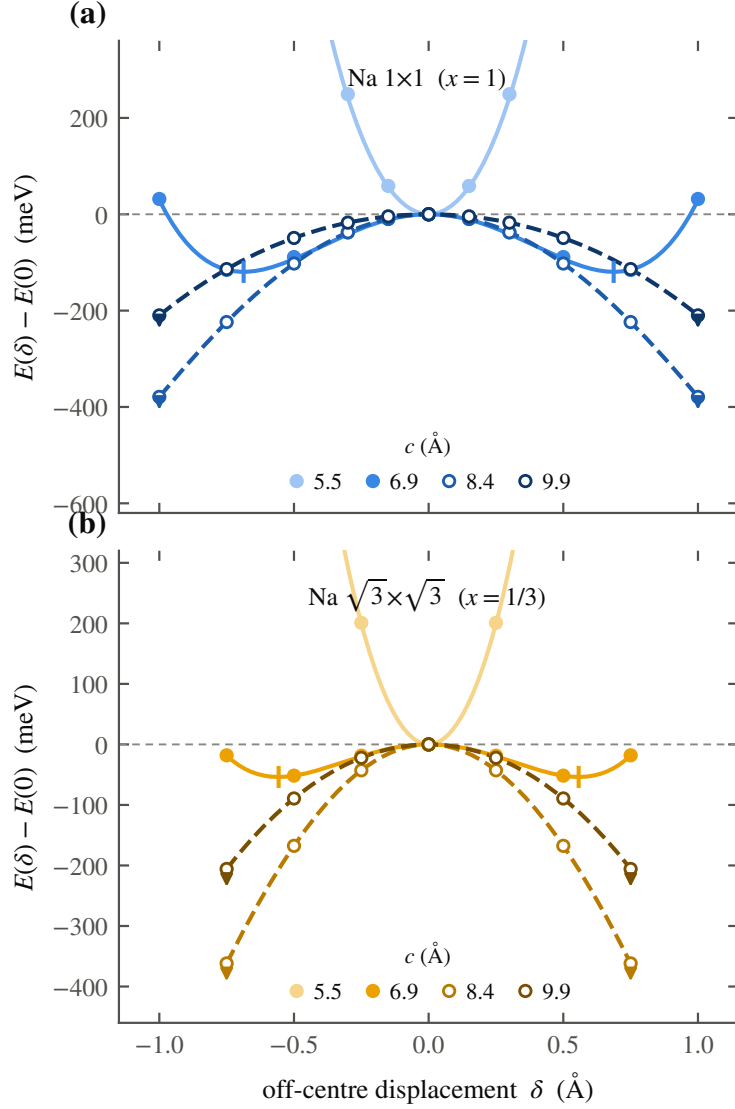


FIG. S1. The single-to-double-well transition. Energy $E(\delta)$ of the off-centre sodium displacement, computed for $\text{Na } 1 \times 1$ (top, $x = 1$) and $\text{Na } \sqrt{3} \times \sqrt{3}$ (bottom, $x = 1/3$) at the four gallery spacings; points are total energies, curves the fits of Eq. (S1) continued to sixth order where needed. At 5.5 Å the well is single and stiff ($\alpha > 0$); by 6.9 Å it has split into a double well ($\alpha < 0$) with minima marked. Open symbols with arrows ($c \geq 8.4$ Å) flag scans still falling at the largest sampled displacement, where the sodium is adsorbing onto one layer and the quoted depth is a lower bound only.

S3. THE TWO-BAND MODEL AND THE COUPLING

The Letter states the pairing formula and defers the derivation to this section: the reason the linear coupling vanishes, the form of the quadratic vertex, and the construction of the dome of Fig. 2(c) from $\lambda(c)$ and $\hbar\Omega(c)$.

The bands. The two states that share the gallery electron are taken from an SSH-type four-band tight-binding model developed for the surfaces of the layered cobaltates [1], rather than fitted here; the model is restricted to the pair of bands relevant in the gallery, one of alkali sp_z character and one of CoO_2 character. Their band centres split according to how the electron is shared between them, and that split is fixed by the Bader charge computed directly in Sec. S2, so the gallery band fills exactly as the two-dimensional gas turns on in Fig. 2(b) of the Letter, with no free parameter left in the electronic part of the model. What is added here is the sodium’s own motion, and how the two bands respond to it. Figure S2 sketches all three pieces of the construction: the charge-sharing bands (panel a), the parity argument (panel b), and the quantized double well with its overtone boson (panel c).

The parity argument. The sodium displacement δ is odd under reflection through the gallery midplane. A linear electron–phonon coupling would shift a band energy in proportion to δ , and such a term must vanish identically at the symmetric point by that reflection alone; we checked this explicitly in the model at several k_z and it holds everywhere in the gallery band. The leading coupling is therefore quadratic, and the band energy responds as $\frac{1}{2}\chi\delta^2$, with

$$\chi = 2 \sum_m \frac{|\langle \text{alk} | \partial H / \partial \delta | m \rangle|^2}{E_{\text{alk}} - E_m}, \quad (\text{S2})$$

the ordinary second-order perturbation sum over the other bands m of the model. Physically, the sum describes the counter-polarization of the ion’s own cloud: the displacement admixes the band of opposite parity, so the gas polarizes against the ionic motion, and the resulting energy gain is even in δ . Near the transition this comes out to $\chi \approx -6 \text{ eV}/\text{\AA}^2$, and it grows in magnitude as the band splitting collapses further out in c .

The overtone boson. Because the coupling is quadratic, an electron scatters off two quanta of the sodium mode rather than one. The matrix element that enters is $\langle 0 | \delta^2 | 2 \rangle$ between the ground state and the second excited state of the quantized well (coupling to the first excited state is forbidden by the same parity selection rule that eliminates the linear term),

and the effective boson is the even overtone $\hbar\Omega = E_2 - E_0$. The dimensionless coupling is

$$\lambda = \frac{2 N(0) g^2}{\hbar\Omega}, \quad g = \frac{1}{2} |\chi| \langle 0 | \delta^2 | 2 \rangle, \quad (\text{S3})$$

with $N(0)$ taken directly from the density-functional calculation of Sec. S2, with no adjustable parameter.

The dome and its boundaries. Inserting $\lambda(c)$, $\hbar\Omega(c)$, and $N(0)(c)$ into the Allen–Dynes formula, with $\mu^* = 0.10$ and gated on $N(0) \geq 0.1$ states $\text{eV}^{-1}\text{cell}^{-1} \text{ spin}^{-1}$, gives the dome shown in Fig. 2(c) of the Letter: a narrow peak between $c \approx 6.35$ and 6.45 Å, height $T_c \approx 14$ K at $\lambda \approx 1.3$ and $\hbar\Omega \approx 11$ meV. The two boundaries mark the limits beyond which the approximation itself ceases to be reliable. Below c^* , $N(0) \rightarrow 0$ and $\lambda \rightarrow 0$ with it, because the gallery band is empty of carriers. Above the window, or once the sodium is frozen into one well, $\hbar\Omega$ stays near 20 meV while λ rises past 2, beyond the weak-to-intermediate-coupling regime and into the self-trapped-polaron regime; Allen–Dynes theory does not describe that regime, and a bipolaron treatment past that cut has not been attempted here. The ansatz range quoted for the peak, 11–23 K, comes from varying the electron–phonon range parameter over its plausible values; the dome height depends on that parameter to the fourth power, which is why the Letter states $T_c \approx 14$ K as an order-of-magnitude estimate rather than a precise value.

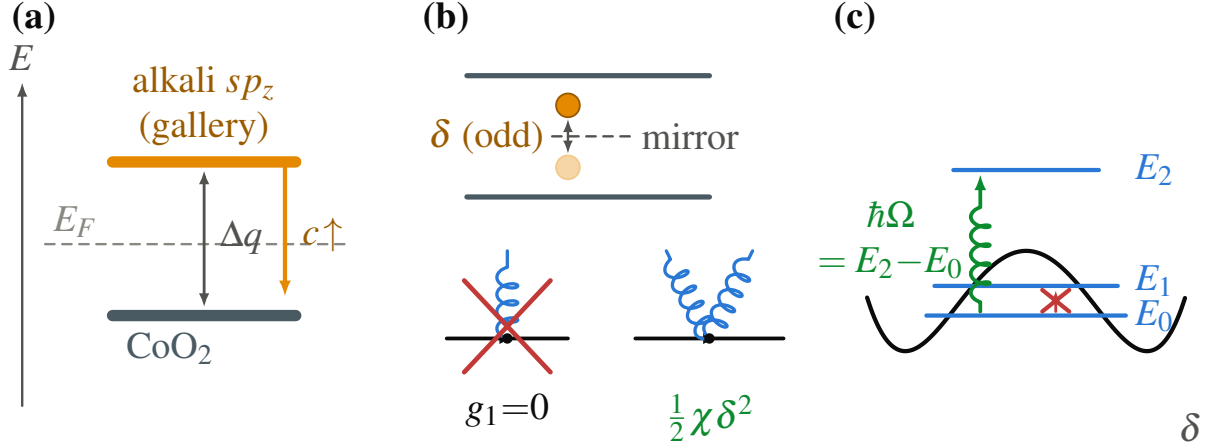


FIG. S2. The two-band model and its coupling. (a) The gallery electron is shared between an alkali sp_z band and a CoO_2 band, taken from the SSH-type surface model [1] and split by the Bader charge computed directly. (b) The gallery midplane mirror forbids a linear electron–phonon vertex at $\delta = 0$; the leading coupling is quadratic. (c) The quantized double well, with the near-degenerate tunnelling doublet E_0, E_1 and the even overtone E_2 that carries the quadratic coupling; the pairing boson is $\hbar\Omega = E_2 - E_0$.

S4. THE FERMI SURFACE AND THE ARPES COMPARISON

The band structure of Sec. S5 shows the gallery band crossing E_F ; this section compares the resulting Fermi surface with experiment. No photoemission exists for the hydrate itself, which dehydrates under the ultrahigh vacuum the measurement requires, so the comparison is necessarily with the anhydrous parent, and its logic deserves a precise statement: it validates a footprint, not a band identity.

Figure S3 is the Fermi surface at $c = 9.9 \text{ \AA}$: the spectral function $A(\mathbf{k}, E_F)$ in the $k_z = 0$ plane of the Na 1×1 hexagonal Brillouin zone, from the dense $24 \times 24 \times 8$ non-self-consistent eigenvalues, symmetry-completed with the C_{6v} point group and Lorentzian-broadened at E_F (50 meV, both spins summed). The calculation gives a single electron barrel around Γ , the sodium two-dimensional gas, and no e'_g pockets at K. Angle-resolved photoemission on the anhydrous parent $\text{Na}_{0.73}\text{CoO}_2$ [2] sees the same topology: a single a_{1g} barrel around Γ with measured $k_F \approx 0.65\text{--}0.70 \text{ \AA}^{-1}$, and no e'_g pockets, the e'_g band sitting $\approx 85 \text{ meV}$ below E_F . Our barrel is sodium- rather than cobalt-derived and the composition differs,

so the comparison concerns topology and size rather than band identity; the two barrels are of comparable radius. In the present picture, then, hydration exchanges the orbital character of the barrel, cobalt for sodium, while nearly preserving its size, and a topology-level measurement cannot distinguish the two. The comparison establishes two facts: the e'_g pockets required by several in-sheet pairing proposals are absent already in the parent, and the discriminating test of the gallery band is a character-resolved measurement, the light-polarization or photon-energy dependence of the barrel's spectral weight on freshly hydrated crystals, which should find sodium rather than cobalt weight. The measured a_{1g} velocities, $v_F \approx 0.3$ and $0.6 \text{ eV \AA}$ along Γ -K and Γ -M respectively, lie roughly a factor of two below the band-theory value, a renormalization not captured by the self-energy-free Kohn-Sham map (see also Sec. S10).

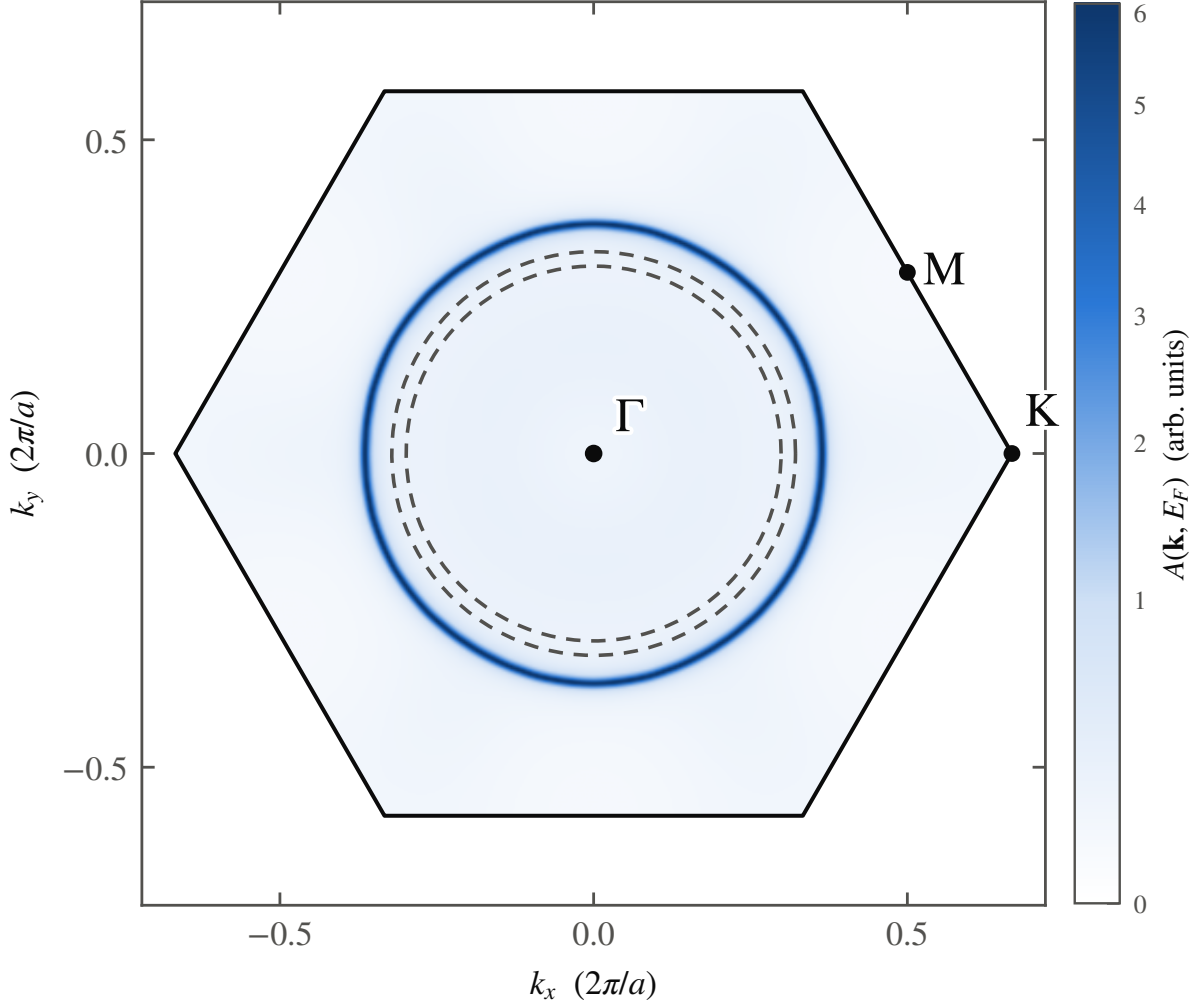


FIG. S3. Fermi surface at $c = 9.9 \text{ \AA}$ and the ARPES comparison. $A(\mathbf{k}, E_F)$ in the $k_z = 0$ plane of the Na 1×1 hexagonal Brillouin zone, symmetry-completed and Lorentzian-broadened (50 meV, both spins summed): a single electron barrel around Γ and no e'_g pockets at K. Dashed: the ARPES barrel on anhydrous $\text{Na}_{0.73}\text{CoO}_2$ [2] ($k_F \approx 0.65\text{--}0.70 \text{ \AA}^{-1}$), of comparable radius; the measured a_{1g} velocities are $\sim 2\times$ below the unrenormalized band value shown here.

S5. ELECTRONIC EVIDENCE FOR THE GALLERY BAND

A. At full sodium coverage

Figure S5 (reproduced from the Letter's Fig. 3, with the full caption repeated here for completeness) shows the band structure from which the Letter's central number, 91% sodium weight at E_F by $c = 9.9 \text{ \AA}$, is obtained. It is bare Kohn–Sham spectral weight, both spins

summed and each band Lorentzian-broadened by $\Gamma = 40$ meV, with no self-energy applied. The cell is weakly spin-polarized above the transition ($0.8 \mu_B$ at $9.9 \text{ \AA}$), so the gallery band is genuinely spin-split; summing both spins is what a photoemission measurement would also collect.

The Bader charge on the sodium tracks the same filling. In the 1×1 cell it grows monotonically from essentially zero at $c = 5.5 \text{ \AA}$ to $0.58 e$ at $9.9 \text{ \AA}$, the number quoted in the Letter. Converted with the in-plane cell area ($a = 2.888 \text{ \AA}$, $A = 7.22 \text{ \AA}^2$), the fully returned charge corresponds to an areal density $n_{2D} \approx 8 \times 10^{14} \text{ cm}^{-2}$; if that charge occupied a single spin-degenerate band, its Fermi radius would be $k_F = \sqrt{2\pi n_{2D}} \approx 0.71 \text{ \AA}^{-1}$, of the same size as the Fermi-surface barrel computed in Sec. S4 and as the measured barrel on the anhydrous parent ($k_F \approx 0.65\text{--}0.70 \text{ \AA}^{-1}$). The Bader count, the Fermi-surface volume, and the factor of ≈ 30 rise of $N(0)$ between 5.5 and $6.9 \text{ \AA}$ [Fig. 2(b) of the Letter] are three mutually consistent measures of one band filling. The same quantity at the physical composition is smaller: in the $x = 1/3$ cell the sodium Bader charge rises by $\Delta q \approx 0.18 e$ between the monolayer and bilayer spacings, the value that sets the valence split of Sec. S12.

B. At the physical composition, $x = 1/3$

The full-coverage result is checked at the real sodium content in the $\sqrt{3} \times \sqrt{3}$ cell. Figure S4 mirrors the $x = 1$ case exactly: no sodium weight at E_F at $5.5 \text{ \AA}$, a strong sodium band through E_F at $9.9 \text{ \AA}$. These are supercell bands, each a fold of the primitive zone; because the plane-wave coefficients required for a weight-resolved unfolding were not retained, we indicate the folding relations rather than construct an unfolded dispersion. The 1×1 K point folds onto the supercell Γ , and the 1×1 M point folds onto the supercell M. Every band in Fig. S4 should be read as folded.

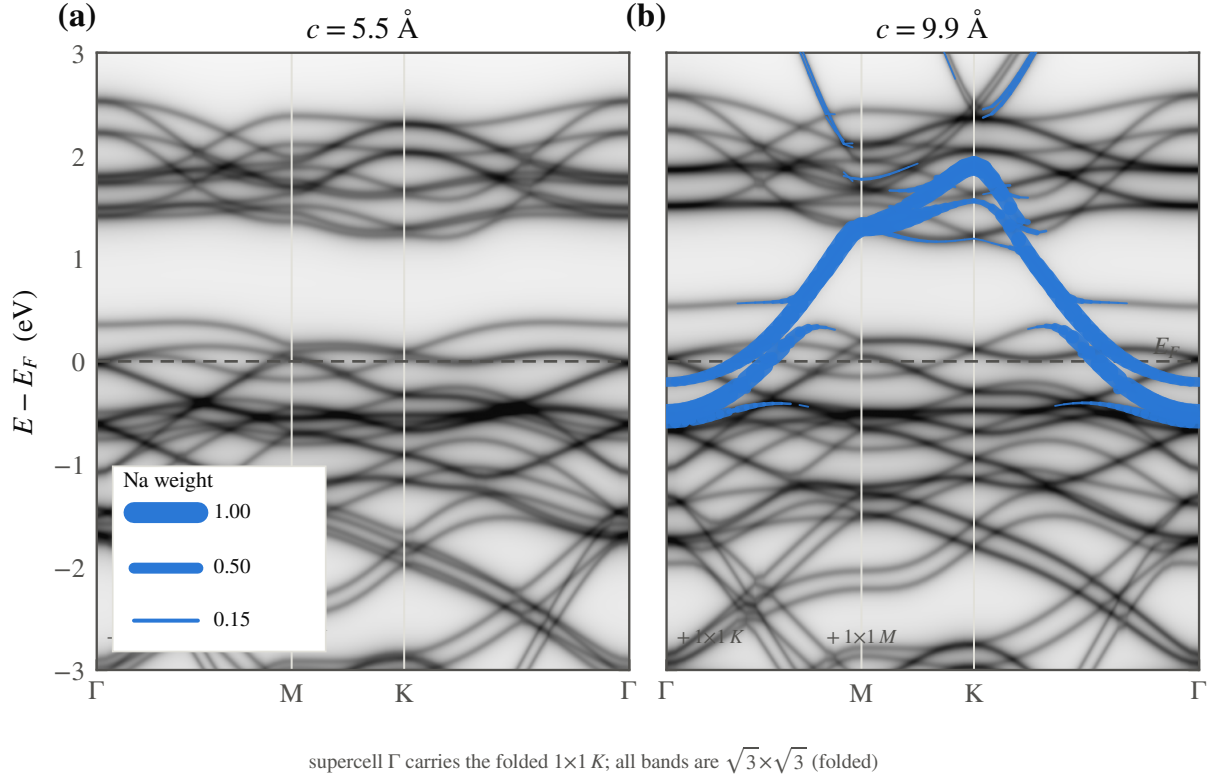


FIG. S4. The gallery band at the physical composition, $x = 1/3$. Kohn-Sham spectral weight of the Na $\sqrt{3} \times \sqrt{3}$ cell at (a) 5.5 and (b) 9.9 Å, built exactly as in Fig. S5 (grayscale total weight, blue sodium fatband, both spins summed, 40 meV broadening). No sodium weight sits at E_F at 5.5 Å; a strong sodium band crosses E_F at 9.9 Å, mirroring the $x = 1$ result. The 1×1 K point folds onto the supercell Γ and the 1×1 M point onto the supercell M; every band shown here is folded.

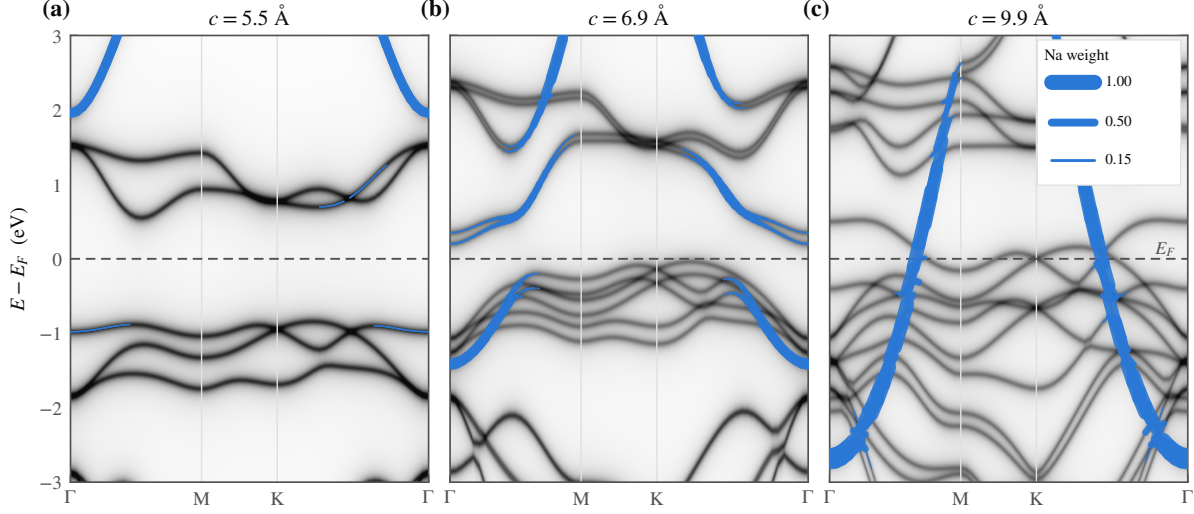


FIG. S5. (Reproduced from the Letter.) Emergence of the gallery band. Kohn-Sham spectral weight of Na 1×1 along Γ -M-K- Γ at (a) 5.5, (b) 6.9, and (c) 9.9 Å, referenced to E_F (dashed). Grayscale is the total DFT spectral weight (all bands, both spins summed, each Lorentzian-broadened by $\Gamma = 40$ meV); the blue fatband is the sodium-orbital projected weight from PROJWFC, its width proportional to sodium character (key in panel c). At 5.5 Å the $x = 1$ cell is a band insulator and the sodium band lies ≈ 2 eV above E_F ; by 9.9 Å a band of $\sim 91\%$ sodium character has descended through E_F , the interlayer electron gas appearing as a single band. No self-energy is applied; this is bare Kohn-Sham weight rather than a renormalized quasiparticle spectrum.

S6. MODE SOFTENING AND THE QUANTIZED WELLS

The pairing model of Sec. S3 requires a soft anharmonic mode, beyond a sign change in α ; this section quantifies the softening.

Figure S6 is the effective zero-point frequency $\hbar\omega_{\text{eff}} = \hbar^2/2M\langle\delta^2\rangle_0$ of the quantized alkali mode, obtained from an exact one-dimensional solution of the fitted wells of Sec. S2 at the physical sodium mass. It collapses by three orders of magnitude across the transition, from a stiff ~ 30 meV single-well mode to a sub-meV soft mode as the gallery opens, entering a quantum-paraelectric regime in which the ion is smeared across the barrier rather than localized in either minimum. The inset shows the lowest levels of the 6.9 Å sodium double well directly: the near-degenerate tunnelling doublet E_0, E_1 that the parity argument of Sec. S3 forbids from coupling linearly, and the even overtone E_2 that carries the quadratic vertex and sets $\hbar\Omega$.

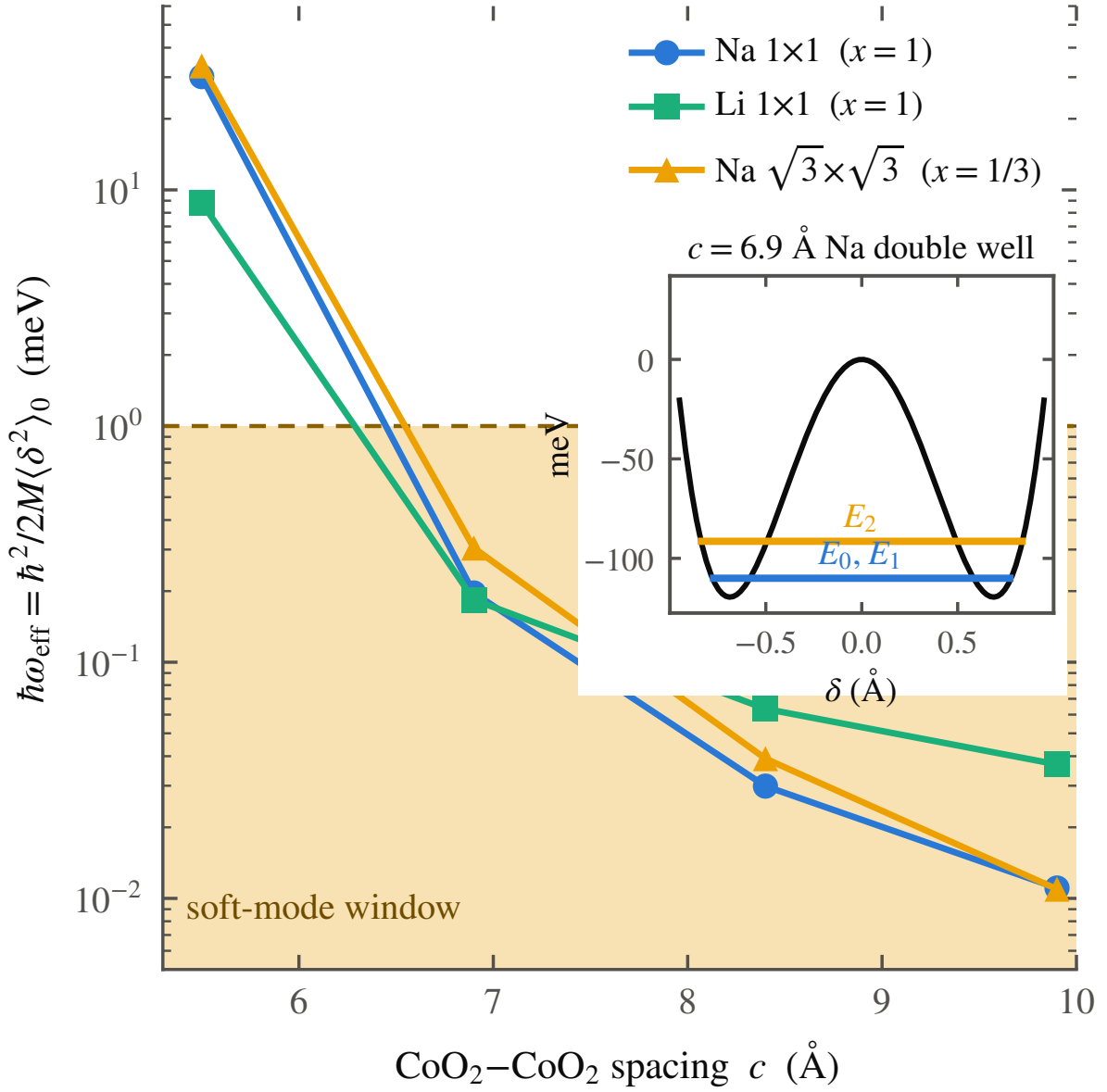


FIG. S6. The sodium mode softening. Effective zero-point frequency $\hbar\omega_{\text{eff}} = \hbar^2/2M\langle\delta^2\rangle_0$ of the quantized alkali mode, from an exact one-dimensional solution of the fitted wells at the physical mass. It collapses by three orders of magnitude across the transition, from a stiff ~ 30 meV single-well mode to a sub-meV soft mode, entering a quantum-paraelectric window. Inset: the lowest levels of the 6.9 Å double well, the tunnelling doublet E_0, E_1 and the overtone E_2 that carries the quadratic coupling.

S7. THE WATER CAGE

The Letter’s Fig. 4 rests entirely on the construction described here. We describe it in full because the result depends on the construction, and because the three-way comparison (vacuum, rigid cage, adiabatic cage) converts a qualitative scenario into a testable calculation.

A. Construction

The host is the $\sqrt{3} \times \sqrt{3}$ ($x = 1/3$) Na_xCoO_2 cell at the bilayer spacing $c = 9.9$ Å ($a = 5.00$ Å), the same cell and the same spin-polarized PBE+PAW settings of Sec. S1 used for the anhydrous scans. The CoO_2 sheets and oxygen heights are frozen at the values used throughout, so the water molecules are the only addition. Each water is a rigid gas-phase molecule, O–H bond 0.9572 Å and H–O–H angle 104.52°. Four are placed per cell, in the arrangement neutron diffraction finds for the bilayer hydrate [3, 4]: the sodium sits on the Co-top (0,0) site on its plane at $z = c/2 + \delta$, and the four water oxygens lie in two layers 1.2 Å below and above that plane, on in-plane sites the sodium does not occupy, two O-top images in the lower layer and two Co-top images in the upper layer. Over the whole displacement scan the sodium–water oxygen distances run 2.05–2.57 Å (lower layer) and 2.92–3.13 Å (upper), an octahedral-like first coordination shell. The hydrogens point away from the sodium, toward the nearest CoO_2 sheet and radially outward, the molecular bisector tilted 50° off the c axis, chosen to maximize the smallest intermolecular contact: every water–water, water–framework, and water–sodium distance stays ≥ 2.05 Å across the whole δ grid. The two down-pointing and two up-pointing molecules cancel the layer’s z -dipole exactly, so the cage adds no net dipole field.

B. The rigid cage (set G)

The waters are held fixed while the sodium scans δ , so $E_{\text{hyd}}(\delta) - E_{\text{vac}}(\delta)$, the hydrate curve minus the identical cell with the twelve water atoms deleted, is the clean effect of the solvation cage on the sodium potential, with no relaxation confounds. This probes the static cage only: it determines whether an expanded gallery leaves the sodium chemisorbed, as in vacuum (an unstable mode with $|\hbar\omega| \approx 11i$ meV), or bound in a confined well, as with

the cage ($\hbar\omega_{\text{eff}} \approx 23$ meV), and it fixes the frequency of that well. It cannot represent the water’s own motion.

C. Adiabatic relaxation (set J)

We next allow the cage to relax. The scan was repeated with the four water molecules relaxed to their minimum-energy configuration at each pinned sodium position, a full BFGS relaxation of the twelve water atoms, one hundred ionic steps, converged at every δ . If the water acted as a lubricant in the naive sense, the relaxing shell would move out of the sodium’s path and flatten the well; instead it follows the sodium off-centre and deepens the bound well, from 199 meV (frozen) to 380 meV at the same $\delta \approx 0.30$ Å minimum, and quantized about that minimum the adiabatic mode is stiffer, near 37 meV, rather than softer. Classical adiabatic water therefore does not soften the sodium potential; taken at face value, this result excludes the naive lubricant picture. A dispersion-corrected (DFT-D3) repeat tracks this deepening at the two points where the D3 trajectory has converged, -210 meV against -222 meV (PBE) at $\delta = 0.15$ Å; the remaining D3 points are still relaxing but are consistent with the same trend, so the deepening is not a van-der-Waals artifact of the bare PBE cage.

D. The 0.4 eV configurational spread and the sixty-step artifact

The single relaxed configuration is only the deepest point of a landscape over which the four-water hydrogen-bond network fluctuates. A relaxation of the same $\delta = 0.30$ Å cell, stopped at sixty ionic steps rather than carried to full convergence, comes to rest 394 meV above the fully converged minimum, a different, shallower local minimum of the cage at the identical sodium position (the open star in the Letter’s Fig. 4(a)). The hydrogen-bond network therefore presents the sodium with a family of potentials rather than a single one, spanning at least 0.4 eV between local minima at fixed δ ; the sodium potential is set by the instantaneous cage configuration, of which the deep adiabatic surface is only the lower envelope. (This also explains why an earlier interpretation of the sixty-step run as a near-flat 5 meV well was an artifact of incomplete relaxation rather than a physical softening; the corrected, fully converged comparison is the one reported throughout.)

E. The timescale argument

The two motions occur on well-separated timescales, and this separation completes the argument. The bound sodium mode has $\hbar\omega \approx 23$ meV, a period near 200 fs; the hydrogen-bond network of gallery water reorganizes on the picosecond scale that quasielastic neutron scattering measures directly [5]. The sodium motion is fast compared with the cage. The ion therefore does not follow the deep adiabatic surface, which the slow network presents only once fully relaxed; it oscillates on a rigid-cage-like surface drawn, at each instant, from the thermal and quantum ensemble of cage configurations, an ensemble that, as the 0.4 eV spread shows, includes surfaces far shallower than the adiabatic minimum. The softening required by this mechanism is therefore statistical-dynamical rather than adiabatic: the fast sodium samples the soft members of the cage ensemble instead of following the slow cage to its stiff, deep minimum. The frozen-cage well is one member of that ensemble and the adiabatic well its extreme deep member; the physical mode lives between them, and the one calculation that can show this directly, a finite-temperature Born–Oppenheimer or path-integral molecular-dynamics sampling of the coupled sodium–water system, remains open.

F. The isotope-null consistency check

This picture makes the deuteration and ^{18}O experiments consistency checks rather than predictions, because both have already been run. Replacing H_2O by D_2O shifts T_c by less than 0.2 K [6]; replacing ^{16}O by ^{18}O is also null [7], with the authors of that study noting explicitly that a null shift does not by itself exclude a phonon mechanism once strong correlations are present. Neither result is evidence against the picture argued here, because neither isotope substitution touches the boson: the pairing mode is the sodium oscillation along c , gated by the statistical ensemble of cage configurations, and the water’s own mass enters only through how stiff the hydrogen-bond network is, a second-order effect on the ensemble the sodium samples rather than a first-order mass term in the boson frequency itself. A mechanism that paired electrons directly to a water vibration would, by contrast, predict a first-order isotope shift and is excluded by this result. Sodium itself offers no equivalent test: ^{23}Na is its only stable isotope, so the direct mass test on the boson is

unavailable, which is why Sec. S12 emphasizes the spectroscopic search for the mode itself over any further isotope substitution.

S8. SPIN PHASE DIAGRAM AND SPIN-ORBIT COUPLING

The gas that condenses in the gallery is not always paramagnetic, and the location of its magnetic order bears on two long-standing puzzles in this compound: the proximity of a magnetic phase to the superconducting dome, and the coexistence of singlet and triplet assignments in the Knight-shift literature. This section constructs the phase diagram that addresses both.

A. The Stoner strip

Where the gas first condenses, the gallery band is narrow and its density of states is high, the conditions for a Stoner instability. The Radha–Lambrecht surface calculation already finds this polarized gallery gas [1], with the gallery moment antiparallel to the cobalt. Carrying the same SSH-type surface-charge model into the gallery and applying a Stoner criterion to the gallery band gives, in the plane of sodium content x and CoO_2 spacing c , a polarized strip that tracks the carrier turn-on line, giving way to a paramagnon-enhanced metal just outside it and to a conventional paramagnet beyond (Fig. S7(a)); the spacing dependence of the enhancement is the cut in Fig. S7(b). The diagram is anchored to the present calculations: the density-functional moment switches on exactly at the well transition (Fig. S7(c)), which is where the model places the carrier turn-on at full coverage. One quantity is excluded as unreliable: the moment of the $\sqrt{3} \times \sqrt{3}$ cell, over-polarized in the local-spin-density approximation and omitted from the diagram; the well energies on which the mechanism rests shift by less than 6% under the spin treatment (Sec. S10), so the exclusion does not affect the structural conclusions.

In this picture the magnetism is a thin strip adjacent to the carrier turn-on, strongest just past it and weakening as the band widens, so that with increasing spacing the turn-on is followed first by the magnetic strip and then by the superconducting region. This is the ordering Sakurai, Ihara, and Takada report for the magnetic phase bordering the dome against cobalt valence [8]. Figure S8 supports this assignment with experimental

data: the measured T_c against cobalt valence, with the superconducting dome and the adjacent magnetic-phase region shown as pastel fills, anchors the adjacency of magnetism and superconductivity in measured data rather than in a schematic. Proximity to the Stoner boundary also resolves a long-standing discrepancy. Early powder NMR suggested a triplet state, whereas later single-crystal work indicated a singlet; in this picture the two results reflect where each sample sits within the strip, because the Stoner enhancement varies steeply across it and hydration drifts from sample to sample, and immediately at the boundary an equal-spin channel of f symmetry lies close in energy to the singlet, connecting to the two f -wave states of Sec. S9 that Mazin and Johannes were unable to eliminate [9]. A specific prediction follows: the NMR Knight-shift suppression below T_c should anticorrelate, across samples, with the normal-state Stoner enhancement. Both symmetry-allowed states of Mazin and Johannes are consistent with the muon null [10], and the chiral $d + id$ scenario [11] is not required to account for the data, though it is not excluded by them.

B. Spin-orbit coupling

Spin-orbit coupling neither selects nor mixes the pairing at any level relevant here. The gallery state is sodium-derived, with only a few percent cobalt weight, so its effective on-band spin-orbit scale is of order 2 meV; the Rashba splitting from the off-centre sodium, estimated from the displaced charge, is about 0.8 meV, of order the gap but far below the Fermi energy, and the parity-mixed component of the gap it induces is at the sub-percent level. On the average centrosymmetric structure the up and down sectors carry opposite Rashba fields and largely cancel. Spin-orbit coupling is therefore a small correction to the spin response and does not enter the pairing.

The computed regimes of the preceding sections assemble into a single phase diagram in the (c, T) plane (Fig. S9). The dome and its walls are computed quantities: the carrier turn-on sits at $c = 5.57 \text{ \AA}$ (the boundary of the no-2DEG region in the coupling scan of Sec. S3), the pairing window closes at $c = 6.43 \text{ \AA}$ where λ reaches 2 and the static description turns polaronic, and between the two the dome peaks at 14 K with its ansatz range. The Stoner strip of this section enters as a bar at $T = 0$: the polarized ground state is established at the sampled spacings, but no ordering temperature has been computed, so the strip is drawn with no temperature extent. The bilayer-hydrate region near $c = 9.8 \text{ \AA}$ is contingent

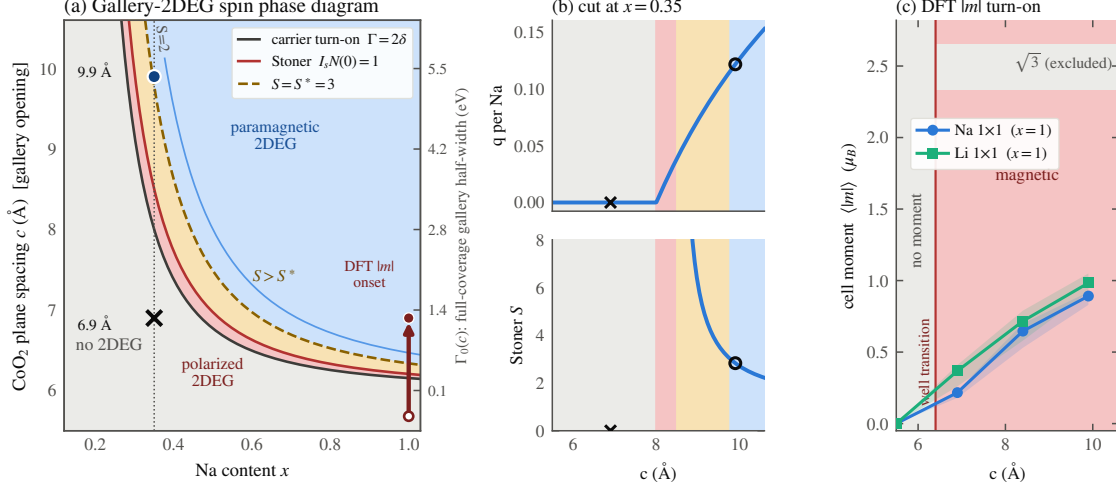


FIG. S7. The gallery-2DEG spin phase diagram. (a) Stoner phase diagram of the interlayer gas, built by carrying the SSH-type surface-charge model of Ref. [1] into the gallery: in the plane of sodium content x and CoO₂ spacing c , the gallery band is empty below the carrier turn-on line, narrow and Stoner-polarized (red) just above it, then a paramagnon-booster (amber) and plain paramagnetic (blue) metal further out. Markers: the monolayer (6.9 Å, $\times$) and bilayer (9.9 Å, $\circ$) hydrates at the physical $x = 0.35$. (b) Cuts at $x = 0.35$: the carrier fraction q per Na and the Stoner enhancement S against c . (c) The DFT cell-integrated moment $\langle |m| \rangle$ for Na and Li 1×1 , switching on at the same spacing the well transition does; the $\sqrt{3} \times \sqrt{3}$ cell is over-polarized in LSDA and shown only as an excluded caveat band. The model is an ansatz calibrated to the one on/off hydrate anchor pair; every parameter is source-tagged in the generating script.

rather than computed: the static theory there is polaronic, and superconductivity in that region requires the dynamical softening by the water cage constructed in Sec. S7, which remains an open test. That region carries the experimental points of the measured dome, whose overlapping H₂O and D₂O sets restate the isotope null; the measured T_c of 2–5 K in the right segment should not be read against the computed 14 K dome in the left segment, which is not a fit to those points.

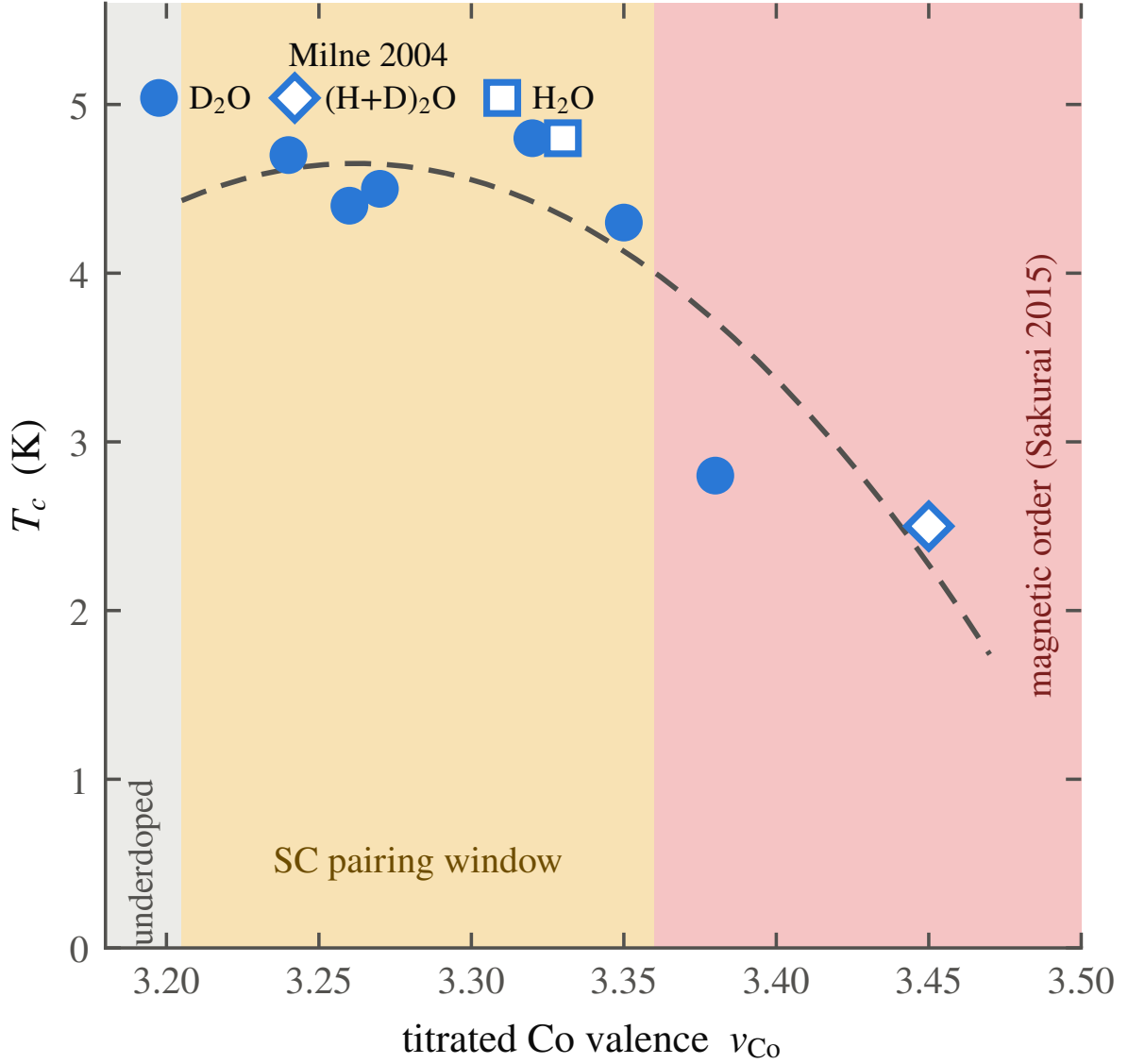


FIG. S8. Experimental evidence that magnetism borders the dome. Experimental T_c against the redox-titrated cobalt valence, the eight batches of Milne *et al.* [12], with the superconducting dome and the adjacent magnetic-phase region drawn as pastel fills; the nominal-composition $(4-x)$ points of Ref. [13] use a different, offset (≈ 0.2 – 0.3) valence axis and are deliberately omitted rather than mixed with the titrated set. The magnetic-phase boundary ($v \approx 3.36$) is anchored to the collapse threshold Milne *et al.* report, combined with the phase adjacency Sakurai *et al.* give [8], and is approximate. The titrated batches cover the optimal plateau and the overdoped collapse but not the underdoped rising edge, so the figure is a partial map of the dome rather than a determination of its full shape; it establishes that the magnetic region sits immediately outside the superconducting one, the adjacency the Stoner strip of Fig. S7(a) predicts from the calculation alone.

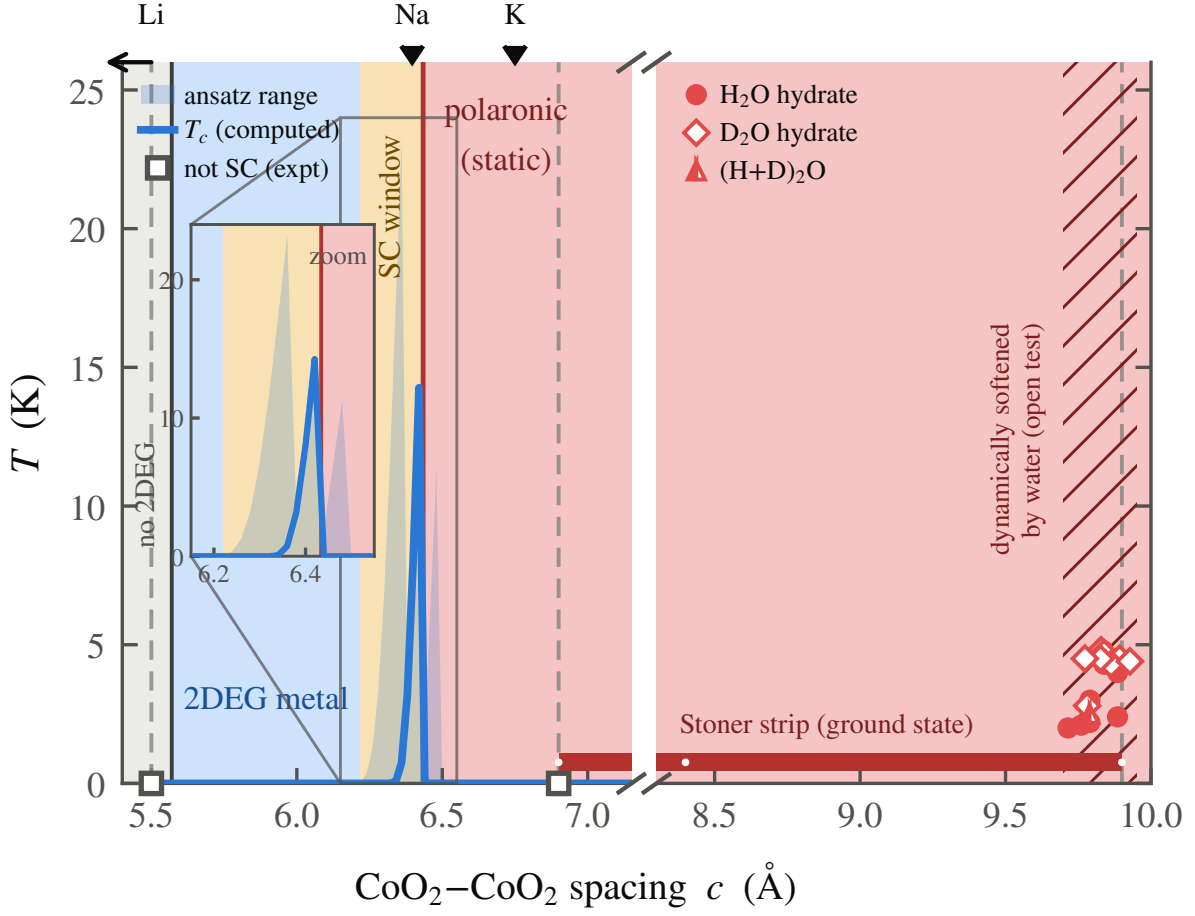


FIG. S9. The assembled phase diagram in the (c, T) plane, broken spacing axis. Computed elements: the carrier turn-on at $c = 5.57$ Å, the T_c dome with its ansatz range (inset: the same dome over its natural 0.4 Å width), and the polaronic wall at $c = 6.43$ Å where $\lambda = 2$, all from the coupling scan of Sec. S3; the alkali thresholds $c_{\text{Li}}^* \leq 5.5$ Å, $c_{\text{Na}}^* = 6.40$ Å, and $c_{\text{K}}^* \approx 6.75$ Å (Sec. S12) are marked on the upper edge. Ground-state-only element: the Stoner bar marks the spacings where the LSDA moment exceeds $0.1\mu_B$; the moment is sampled only at 6.9, 8.4, and 9.9 Å (white dots), so the bar spans the sampled extent, and it carries no temperature extent because no ordering temperature is computed. Contingent element: the hatched sub-band near $c = 9.8$ Å marks where superconductivity requires the dynamical softening by the water cage (Sec. S7), an open test; no computed T_c curve is drawn in the right segment. The points in that sub-band are the measured bilayer-hydrate dome ($T_c = 2\text{--}4.8$ K), with H_2O and D_2O samples overlapping (the isotope null); they are measurements, not points on the computed dome. Open squares on the $T = 0$ line: the anhydrous (5.5 Å) and monolayer-hydrate (6.9 Å) compounds, which do not superconduct.

S9. EXPERIMENTAL CONSTRAINTS ON THE PAIRING STATE

Proposals for the pairing state of $\text{Na}_x\text{CoO}_2 \cdot y\text{H}_2\text{O}$ advanced over two decades are constrained by four key measurements; Table S2 evaluates the principal proposals, including the state advocated in this work, against each of them.

The natural ground state of a phonon-mediated instability on the gallery gas is a spin singlet, anisotropic and $s_{\pm}$ -like across the two bands, the gallery two-dimensional gas and the a_{1g} pocket, and it breaks no time-reversal symmetry. These properties alone satisfy the two most restrictive constraints in Table S2: the singlet is consistent with the single-crystal Knight shift, which excludes the triplet proposals, and the preserved time-reversal symmetry is consistent with the μSR null result [10], which excludes the chiral $d + id$ states [11].

The remaining two constraints are accounted for naturally within the present picture. There is no Hebel–Slichter peak because the pairing boson is a soft, strongly anharmonic mode, and the anharmonic broadening that softens it also smears the coherence peak; we note this as an expected consequence of soft-mode coupling rather than as a fitted result. The specific-heat nodes are the more telling case. The gap on the gallery band is small, set by the dilute gallery density of states, and a small gap on a two-dimensional gas is fragile against precisely the gallery disorder that dehydration introduces. A fragile second gap suppressed below the measurement threshold mimics line nodes, which explains both the node claims and their reported dependence on sample age [8]: the gap carries a fragile second component, removed by the aging of the water network, rather than true nodes. The proposed state is falsifiable in two ways: any observation of broken time-reversal symmetry, a μSR relaxation onset at T_c or a polar Kerr rotation, would exclude the singlet channel, as would a Knight shift that remains constant below T_c in a fresh single crystal, because a singlet requires the shift to decrease. Either observation would exclude the proposed state.

A. The record in detail

Table S3 expands Table I of the Letter: each observation, its source, and the line of reasoning by which the mechanism accounts for it.

TABLE S2. Experimental constraints on proposed pairing states. Verdicts are single words read against each state’s own symmetry: pass, fail, or tension (a constraint met only with an added assumption, flagged ^{a-f}, notes below). The Sakurai entry is a mechanism, not a fixed symmetry, so the symmetry columns do not constrain it. The last row is the state argued in the Letter (Sec. S9).

Proposed state	Knight shift	Hebel– Slichter	Heat nodes	μ SR TRS
RVB singlet $d + id$ [14–16]	pass	pass	tension ^a	fail
Spin-triplet p/f [17, 18]	fail	pass	pass	pass ^b
Chiral $d + id$, fRG [11]	pass	tension ^a	tension ^a	fail
Two surviving f -wave [9]	tension ^c	pass	pass	pass
s -wave, anharmonic H/O phonons [4]	pass	fail ^d	fail ^e	pass
Magnetic-fluctuation [8]	—	—	—	—
This work: singlet $s_{\pm}$, TRS-preserving, two-gap	pass	pass ^f	pass ^g	pass

^a Fully gapped states must invoke disorder or a second gap to explain the observed T^2 term. ^b Requires e'_g pockets that ARPES does not see [2]. ^c Passes only after an added reconciliation of the singlet-favoring Knight shift with a triplet order parameter. ^d Predicts a Hebel–Slichter peak the data do not show. ^e Fully gapped; does not reproduce the observed low- T nodal-like term. ^f Anharmonic broadening of the soft-mode boson smears the coherence peak. ^g Small gallery gap is disorder-fragile; explains both the node claims and their sample-age dependence.

TABLE S3. Detailed version of Table I of the Letter: the mechanism against the experimental record. A verdict of “explained” follows from the static calculation; “passed” is a pre-existing null result the mechanism requires; “matched” is a computed quantity against a measured one; the bilayer-hydrate entry is contingent on the dynamical water softening of Sec. S7.

Observation	Verdict	Source	Why it follows
Anhydrous (5.5 Å) not SC	explained	[19]	$\alpha > 0$: a single stiff well (~ 30 meV) with $N(0) \approx 0$, no gallery gas.
Monolayer hydrate (6.9 Å) not SC	explained	[19]	Beyond the window, $\lambda \approx 20 \gg 2$: the sodium self-traps into a polaron.
Bilayer hydrate (9.9 Å) SC at 4.5 K	explained*	[20]	Static geometry is polaronic; water rebinds the Na and adiabatic relaxation deepens the well; softening into the window is statistical, the open dynamical test (Sec. S7).
T_c is a few K, not tens of K	explained	[20]	Allen–Dynes with the DFT ω_{eff} and λ peaks at 14 K (range 11–23 K), the correct order of magnitude.
No superconducting Li analogue	explained	—	$c_{\text{Li}}^* \leq 5.5$ Å and 8.8 meV zero-point energy keep the light ion centred.
Magnetic phase borders the dome	explained	[8]	The local moment turns on at the same spacing as the well transition (Sec. S8).
Singlet–triplet NMR conflict	explained	[8]	Powder and single-crystal samples sit on opposite sides of the Stoner boundary (Sec. S8).
H/D and $^{16}\text{O}/^{18}\text{O}$ nulls	passed	[6, 7]	The pairing boson is the sodium mode; the water mass enters only through the anharmonicity of the confinement.
Γ -barrel footprint, no e'_g pockets	matched	[2]	Measured on the anhydrous parent (the hydrate dehydrates in vacuum). The gallery barrel reproduces the footprint ($k_F \approx 0.7 \text{ Å}^{-1}$) with sodium rather than cobalt character; character-resolved photoemission on the hydrate is the discriminating test (Secs. S4 and S5).

S10. CORRELATION EFFECTS BEYOND THE PRESENT TREATMENT

Every estimate above is built on Kohn–Sham bands, and the cobaltates are correlated; this section states the direction in which correlations shift each estimate.

Angle-resolved photoemission on the anhydrous parent finds dispersions narrower than the LDA bands by roughly a factor of two, with a_{1g} velocities $v_F \approx 0.3\text{--}0.6$ eV Å against a band value about twice larger [2]. A mass renormalization $m^*/m \approx 2$ raises the density of states $N(0)$ and lowers v_F in the same proportion, and $N(0)$ enters the coupling in the direction that increases it: since $\lambda = 2N(0)g^2/\hbar\Omega$ was built on the unrenormalized density-functional $N(0)$, the λ and T_c estimates of Sec. S3 are, if anything, conservative. The same flattening deepens the narrow-band limit the gallery gas already sits in, and a flatter band at fixed filling carries a higher $N(0)$, which strengthens the Stoner-strip argument of Sec. S8: correlations widen the polarized region of Fig. S7 rather than narrowing it. The natural next step is a DFT+DMFT calculation on the gallery band, which we have not attempted. One caution acts in the opposite direction: in addition to renormalizing the band, correlations can broaden and damp the sodium mode that carries the pairing, and a strongly overdamped mode would couple less efficiently than the sharp Einstein boson assumed here.

The remaining approximations are summarized briefly. The transition spacing is computed at full sodium coverage, $x = 1$, and checked at the physical $x = 1/3$ in a single cell, where it moves by about 0.2 Å. The local-spin-density treatment over-polarizes the $\sqrt{3}$ cell, giving an unphysically large and flat moment there; we base the magnetic analysis on the 1×1 series and exclude the $\sqrt{3}$ moments, and the well energies themselves are insensitive to the spin treatment at the few-percent level, so the over-polarization does not affect the wells. The explicit water (Sec. S7) is treated statically, a rigid cage frozen while the sodium scans and an adiabatically relaxed cage at each sodium position; together they settle the static question and show that adiabatic relaxation deepens rather than softens that well, but neither can represent the statistical-dynamical softening the moving network supplies, the one calculation left open. The pairing is treated as a single Einstein boson in the Allen–Dynes approximation, with an explicit cut where the coupling leaves the regime that approximation can describe; a real bipolaron treatment past that cut is not attempted.

S11. THE DESIGN RULE, PRIOR ART, AND THE LIZRNCL NULL HYPOTHESIS

If the picture holds, it constitutes a design rule rather than an account of a single compound; this section states the rule, identifies the ingredients with precedent in the literature, and tests the rule against the one layered intercalate that appears, at first inspection, to violate it.

Each ingredient of the rule has precedent. Pairing tuned to a soft polar mode at a quantum-critical point is established in SrTiO_3 [21–24]; an occupied interlayer state acting as the switch for superconductivity in an intercalate, controlled by the layer separation, is established for the graphite intercalation compounds [25, 26]; enhancement of T_c as an intercalant mode softens, followed by collapse when the ion orders, is observed in CaC_6 under pressure [27]; a rattling alkali ion in an oversized cage driving strong-coupling pairing is realized in the β -pyrochlores [28, 29]; and carriers in one layer paired by an adjacent quantum-paraelectric layer were proposed by Shenoy, Subrahmanyam, and Bishop [30]. What is new here is the combination: the donor ion is itself the quantum-paraelectric unit rather than a caged spectator or a host-lattice mode, and a single structural parameter, the gallery spacing, simultaneously switches on the carriers and tunes the ion through its bistability, so the window closes on both sides rather than one. Stated as a rule: when a layered gallery is opened past the spacing at which the donor ion becomes bistable, the same structural change that transfers the ion’s electrons into an interlayer gas softens the ion into the pairing boson.

A rule stated this broadly must be tested against counterexamples, and the most extensively studied layered donor intercalate provides one. In Li_xZrNCl and its hafnium analogue [31], opening the gallery from 9 to more than 22 Å does not suppress the superconductivity: T_c rises modestly and then saturates (Fig. S10(a)), and it survives with no intercalant ion at all, whether the chlorine is de-intercalated or the carriers are supplied by a gate [32]. This would constitute a refutation only if the premise of the rule applied, and it does not: here the donor’s electron does not return to the gallery. It fills an in-plane Zr/Hf d -N p host band, the ion remains fully ionized with no spectral weight at the Fermi level, and the pairing belongs to the host rather than to any gallery gas or soft ion. The nitride therefore lies outside the scope of the rule; it tests a different pairing channel, and it

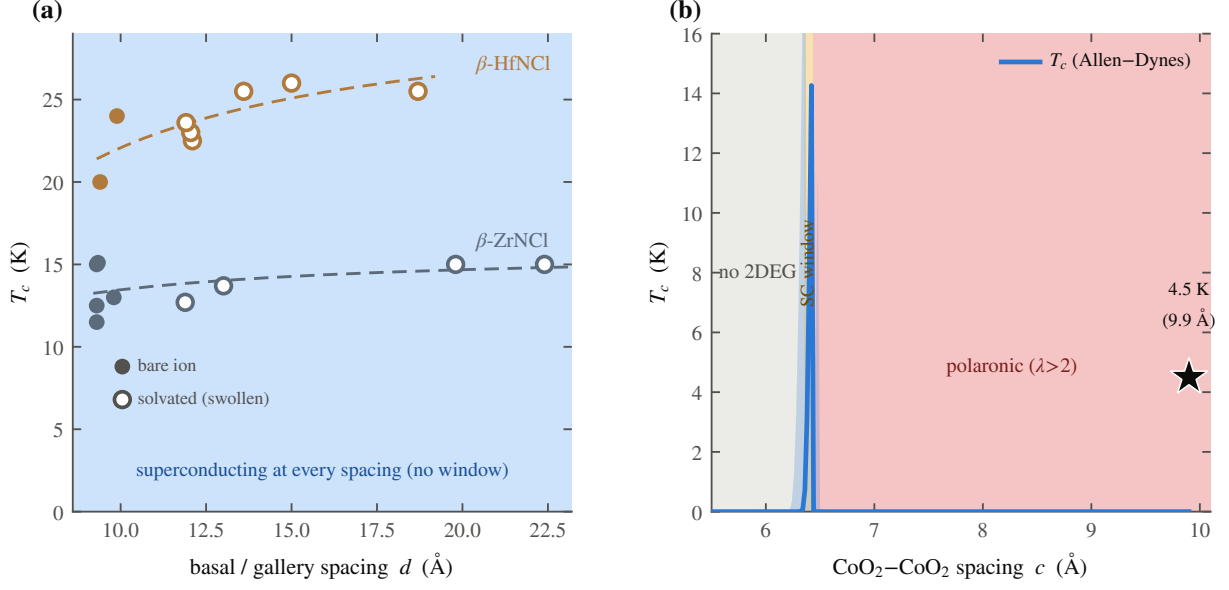


FIG. S10. Test of the null hypothesis. (a) Literature T_c against gallery spacing for Li_xZrNCl and its hafnium analogue [31, 32] (17 points spanning both chemical and gate-induced intercalation): opening the gallery from 9 Å to beyond 22 Å gives a monotonic, saturating $T_c(c)$, because the donor electron never returns to the gallery. (b) Our computed Allen–Dynes dome for the hydrate, with the 4.5 K bilayer-hydrate anchor (star), rises and falls instead: a nonmonotonic window that appears only when the gallery hosts a soft donor ion. The comparison between (a) and (b) defines the null hypothesis against which the present mechanism is tested.

constrains the present proposal in two ways. First, it excludes a universal statement of the rule: the rule holds only where the interlayer state lies at the Fermi level, the precondition made computable by Csányi and co-workers [25], and fails where the host band lies below it. Second, it fixes the null hypothesis: a gallery opened without a soft donor ion gives a monotonic, saturating $T_c(c)$, and the nonmonotonic dome computed for the hydrate (Fig. S10(b)) is, against that background, the anomaly that requires the additional ingredient, the soft sodium ion, absent in the nitride.

S12. PREDICTIONS BEYOND THE COMPOUND

If the rule of Sec. S11 is correct, it carries implications for neighboring compounds and for probes not yet applied to this one; this section collects those predictions and quantifies

them where possible.

A. Lithium, sodium, and potassium

Lithium cobaltate shares the structure with a smaller ion, and superconductivity has never been observed in it; the present picture requires this. Its well crosses into the double-well regime only at the smallest spacing in the dataset, so there is only an upper bound $c_{\text{Li}}^* \leq 5.5 \text{ \AA}$, and at that spacing the well is only 4 meV deep while the lithium zero-point energy is 8.8 meV, larger than the well itself. The light ion does not localize in either minimum; it is smeared across both, a quantum paraelectric, dynamically centred despite the negative α , and a dynamically centred ion provides no soft anharmonic mode. Lithium therefore fails to supply the pairing boson because of its low mass. A heavier gallery ion shifts the threshold in the opposite direction: for potassium the Landau coefficient crosses zero at $c_{\text{K}}^* \approx 6.75 \text{ \AA}$, against 6.40 $\text{\AA}$ for sodium and $\leq 5.5 \text{ \AA}$ for lithium, ordering $\text{Li} < \text{Na} < \text{K}$ with ionic size (Fig. S11); hydrated to that spacing, potassium cobaltate should superconduct by the same route with a heavier, softer mode, a direct prediction for hydrated K_xCoO_2 .

B. The valence signature

The gallery gas produces a signature in the sodium valence that two classes of probe should measure differently. If the sodium keeps a fraction q of its electron in the gallery rather than giving it to the cobalt, a site-specific valence probe, the cobalt L -edge or NQR, and a charge-counting probe, redox titration of the whole crystal, should disagree, and only once the gallery is open. This can be quantified: the Bader charge on the sodium in the $x = 1/3$ cell rises from the empty-gallery monolayer spacing to the superconducting bilayer by $\Delta q \approx 0.18 e$, so the gallery-specific valence split is $\Delta v = x \Delta q \approx 0.06$ per cobalt, switching on with the water and vanishing in the monolayer hydrate where the gallery band is empty. This small splitting is distinct from the large valence offset reported in these materials [12], which requires an additional electron reservoir such as oxonium and is several times larger; the present value is the small, gallery-specific contribution alone. Figure S12 gives the microscopic slope behind the number: the Landau coefficient α against the frozen oxygen height z_O at $c = 9.9 \text{ \AA}$ (vacuum cell, three collinear points: the production $z_O = 0.96 \text{ \AA}$ and

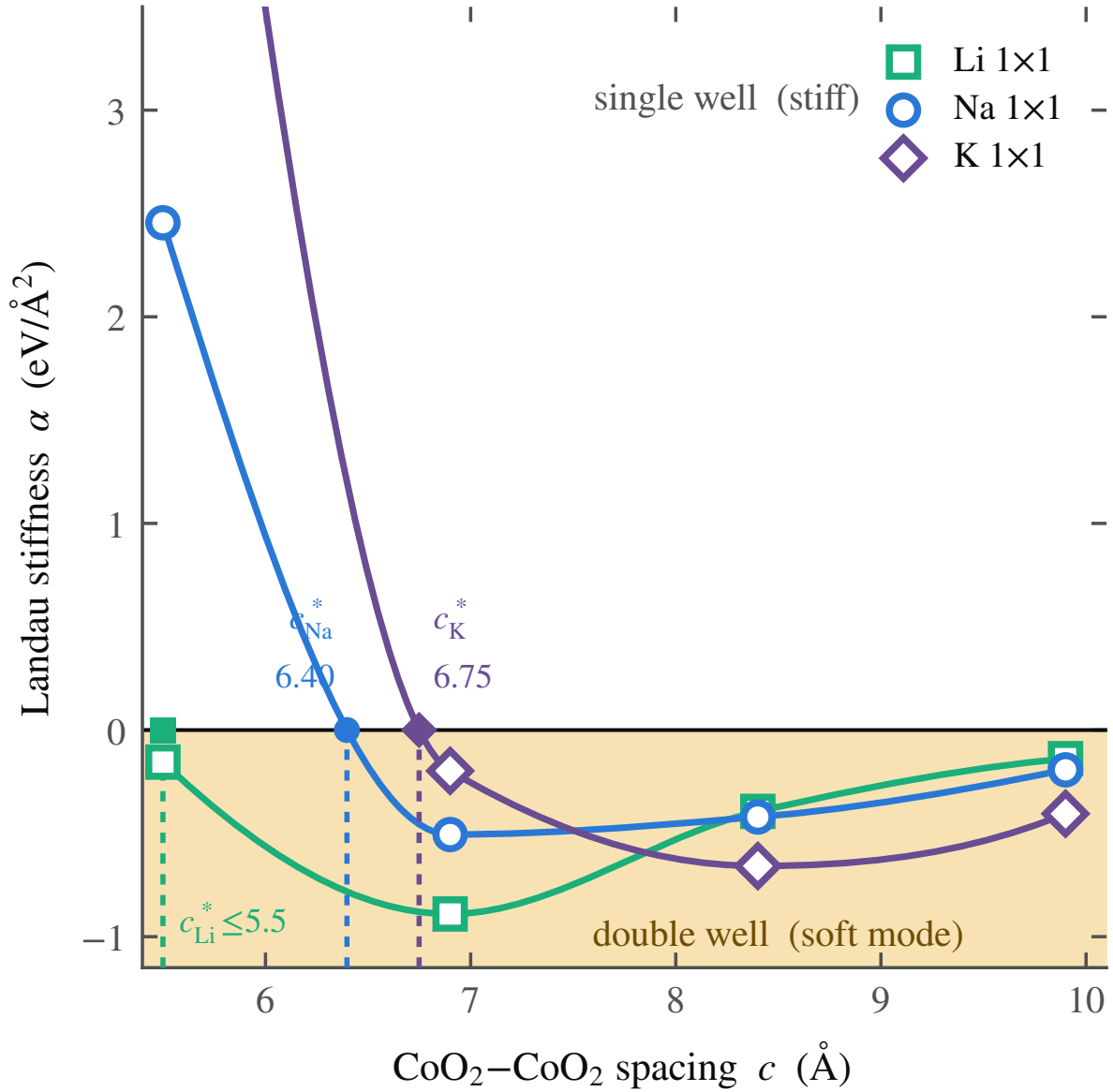


FIG. S11. The threshold ordering. $\alpha(c)$ for Li, Na, and K 1×1 , with the three thresholds marked: $c_{\text{Li}}^* \leq 5.5 \text{ \AA}$ (an upper bound only, α is already negative at the smallest spacing sampled, not a bracketed crossing), $c_{\text{Na}}^* \approx 6.40 \text{ \AA}$, and $c_{\text{K}}^* \approx 6.75 \text{ \AA}$ (both bracketed crossings). The threshold rises with ionic size, the prediction behind the hydrated- K_xCoO_2 test proposed in the Letter.

two dedicated checks at $z_{\text{O}} = 0.90$ and $1.02 \text{ \AA}$) has slope $d\alpha/dz_{\text{O}} \approx -0.52 \text{ eV}/\text{\AA}^2$ per $\text{\AA}$, the same charge-transfer channel that sets Δv , made explicit as a single derivative rather than left implicit in the total-energy scan. Only the slope is defensible here, since three points determine a line but not an absolute stiffness; it is presented as the microscopic rate

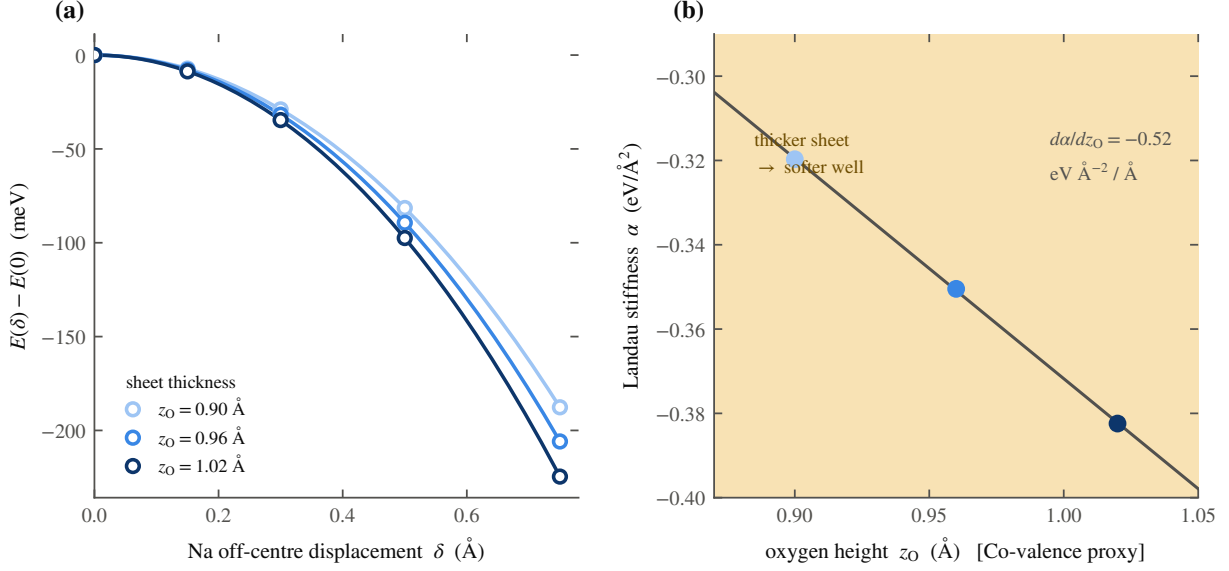


FIG. S12. The microscopic basis of the valence prediction, all vacuum (water-free) potentials at $c = 9.9$ Å. (a) The raw $E(\delta)$ curves at the three sampled oxygen heights $z_O = 0.90, 0.96$ (production), and 1.02 Å, shown for transparency. (b) The extracted Landau coefficient $\alpha(z_O)$: three collinear points give slope $d\alpha/dz_O \approx -0.52$ eV/Å² per Å. The defensible quantity is this slope, not an absolute well stiffness; it is the same oxygen-to-gallery charge-transfer channel that sets the valence split $\Delta v \approx 0.06$ per cobalt quoted in the text.

underlying Δv rather than as a fourth independent calibration of it.

C. Gating and a negative control

A LiCoO₂ thin film in which the carrier density is set electrostatically rather than chemically should show the gallery turn-on without the disadvantage of the light frozen ion. Adding carriers to the anhydrous (5.5 Å) lithium cell by a compensating background, 0.1–0.3 e per formula unit, fills the gallery: the Fermi level and the areal density rise monotonically, to $n_{2D} \approx 4 \times 10^{14}$ cm⁻² at 0.3 e , and the lithium well softens as it fills while remaining under a millielectronvolt deep, far below the lithium zero-point energy of ~ 9 meV, so the ion stays dynamically centred. Beyond its role as a prediction, this calculation serves as a control for the mechanism itself: the charge is added with the lattice fixed at the closed spacing, and the gallery band fills and the well softens nonetheless, demonstrating that the

carrier return, and not the expansion per se, is the agent of both changes (this is the control cited in the Letter). Gating thus tunes the gas without freezing the ion, allowing the gas and the magnetism to be tested independently of the pairing boson; this is a prediction for ionic-liquid-gated thin films.

$\text{Na}_2\text{CoSe}_2\text{O}$ [33], a mixed-anion cobaltate ($T_c = 6.3$ K) that carries the same two-dimensional cobalt network with a different gallery chemistry, provides a test of whether its gallery ion hosts an interlayer band. It does not: the states at the Fermi level are the Co $3d$ –Se $4p$ network, and the ionic sodium of its Na_2O gallery sits well above the Fermi level, with no sodium-derived band there. $\text{Na}_2\text{CoSe}_2\text{O}$ superconducts through its cobalt network rather than through the gallery, and serves as a negative control: the same CoO_2 -like sheet, with no gallery gas, superconducting by a different route.

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